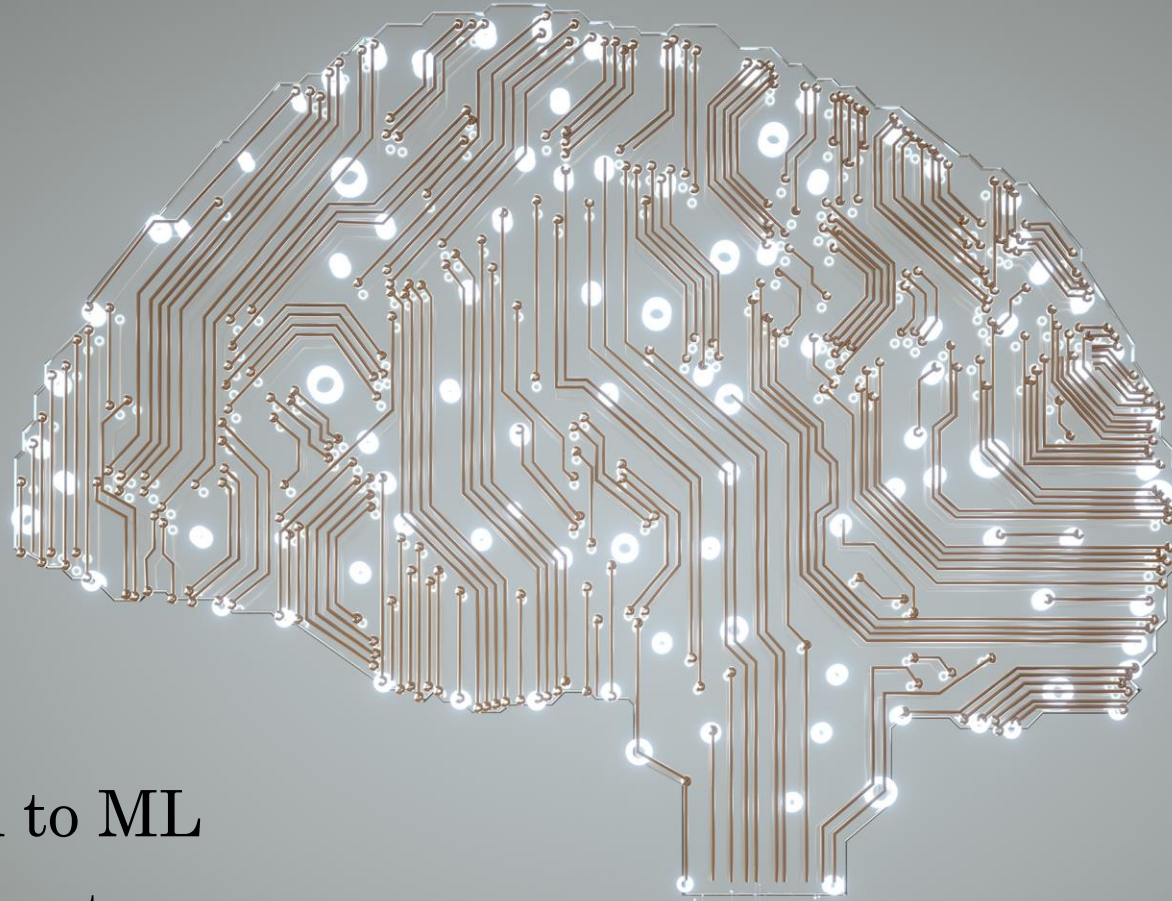
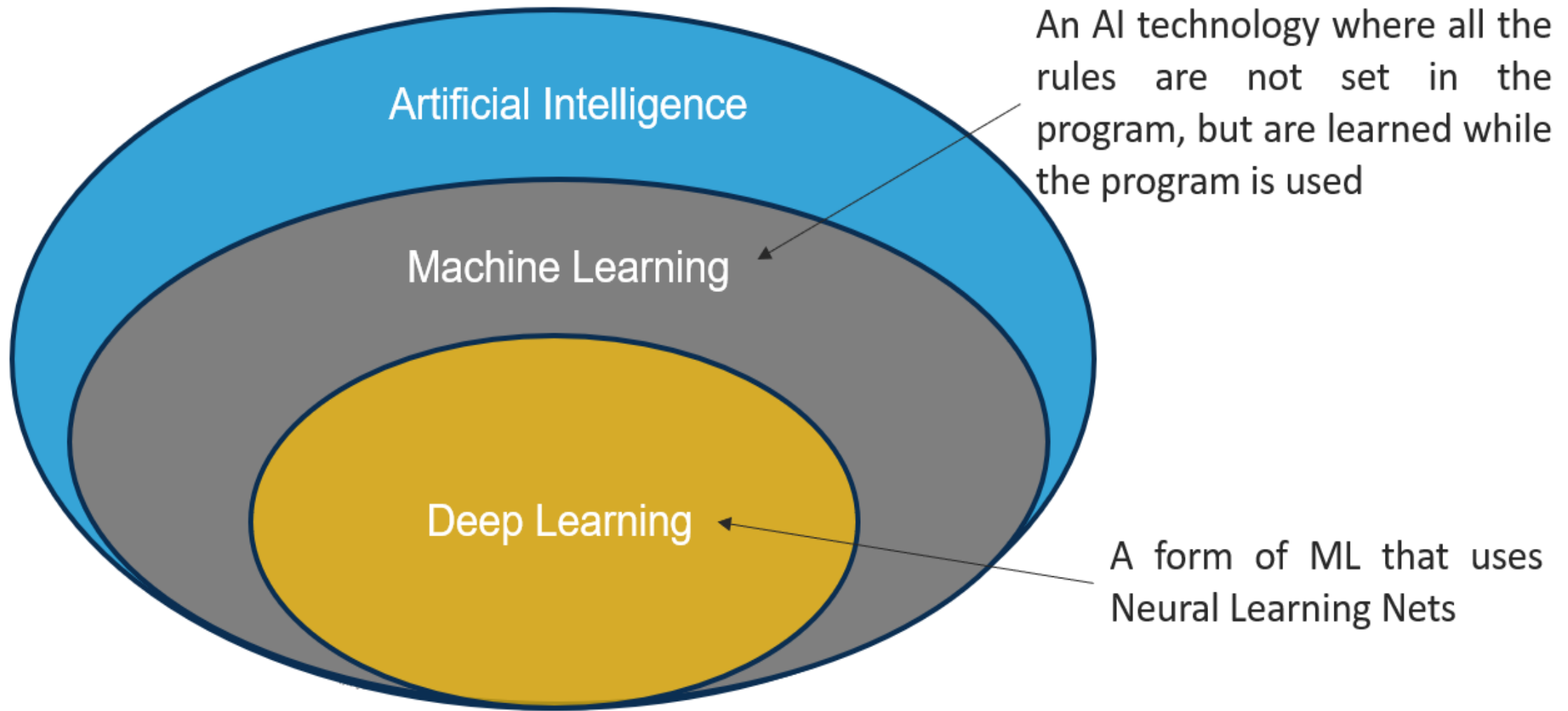


ML & NLP: PGR210



Introduction to ML

Dr. Arvind Keprate



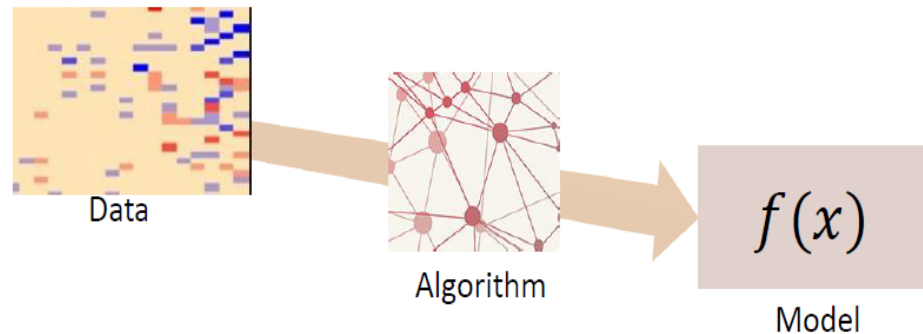
AI - ML - DL?

What is ML?

Machine Learning

Field of AI that gives “computers the ability to learn without being explicitly programmed” - Arthur Samuel

“A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E .”
- Tom M. Mitchell



- Machine learning teaches computers to do what comes naturally to humans and animals: learn from experience.
- Machine learning algorithms use computational methods to “learn” information directly from data without relying on a predetermined equation as a model.
- The algorithms adaptively improve their performance as the number of samples available for learning increases.

What is ML?



- “ **Machine Learning is ...**
an algorithm that can learn from data without relying on rules-based programming.
- Statistical Modelling is ...**
formalization of relationships between variables in the form of mathematical equations.

Statistics vs. Machine Learning

Aspect	Statistics	Machine Learning
Goal	Understand data & relationships	Make predictions from data
Focus	Inference, explanation	Prediction, automation
Assumptions	Strong assumptions (e.g. linearity)	Fewer assumptions; pattern-driven
Techniques	Regression, ANOVA, Hypothesis testing	Neural nets, Trees, SVM
Interpretability	Often interpretable	Often black-box models
Example	Does exercise reduce stress?	Predict stress levels from data

- ✓ Use ****Statistics**** when you want to explain or infer relationships.
- ✓ Use ****Machine Learning**** when prediction accuracy is more important than interpretability.
- 🎯 They often complement each other in modern data science.

Traditional Programming: you code the behavior of the program



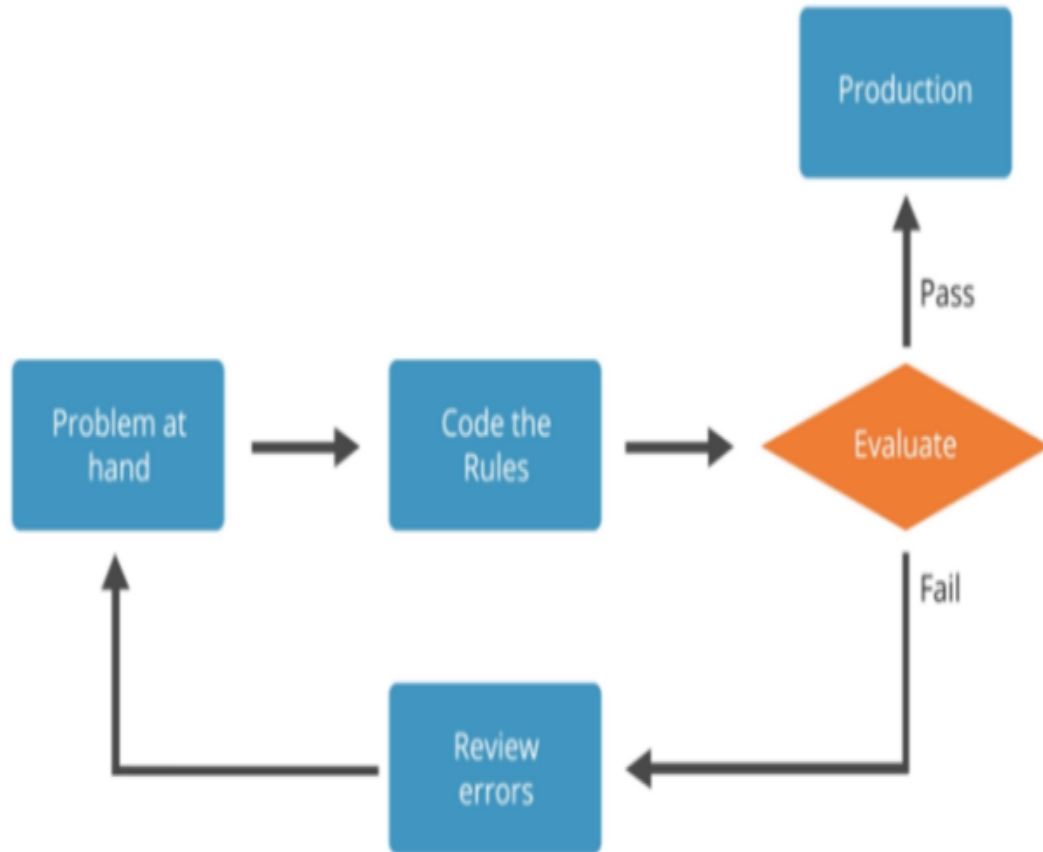
Machine Learning: you leave a lot of that to the machine to learn from data



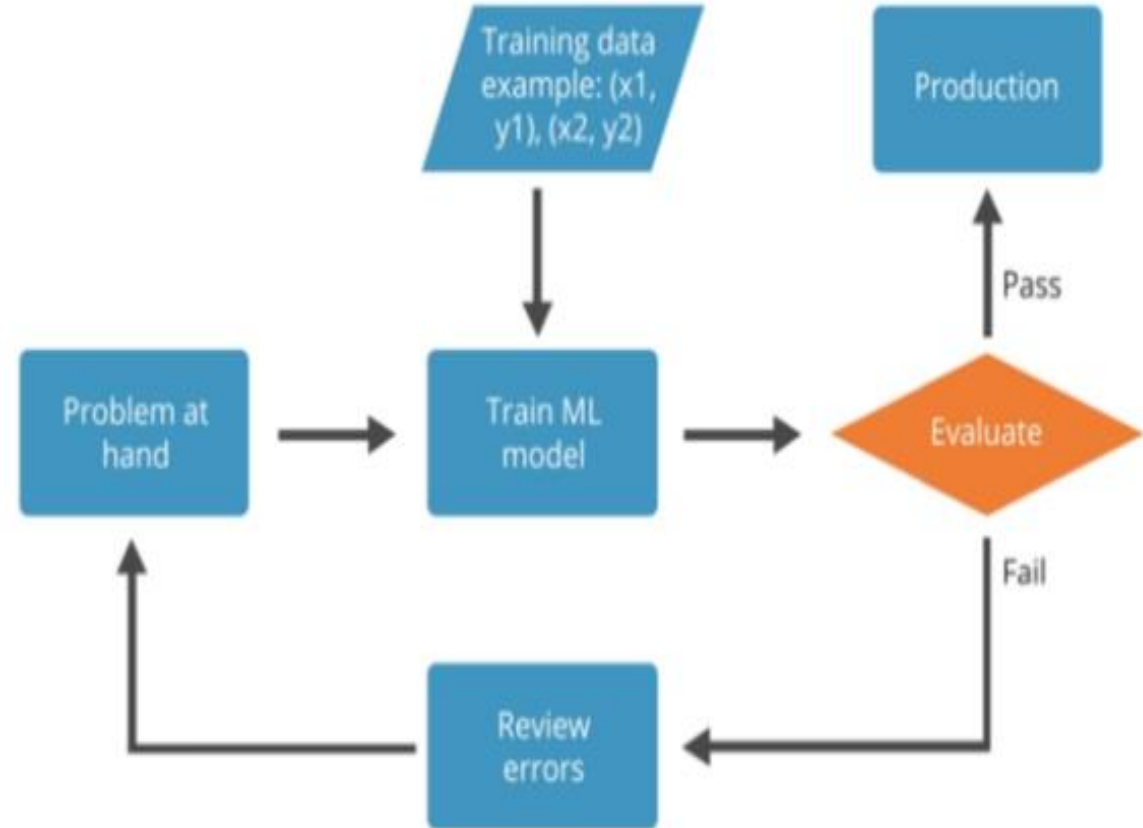
Traditional Programming vs. Machine Learning

Traditional Programming vs. Machine Learning

Traditional programming relies on hard-coded rules.



Machine learning relies on learning patterns based on sample data.



Traditional Programming vs. Machine Learning

◆ Traditional Programming Use Case: Validating a Password

- Rules: At least 8 characters, 1 number, 1 special character
- Easy to write using if-else statements
- ✓ Best when rules are clear and fixed

🤖 ML Use Case: Classifying Emails as Spam or Not

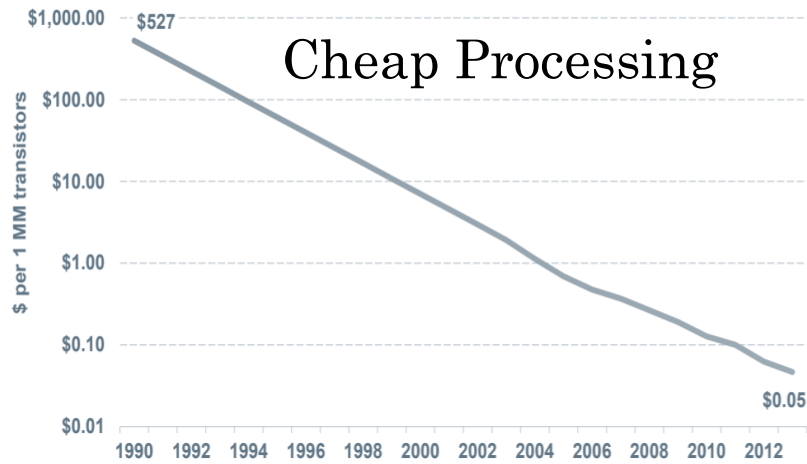
- Input: Subject line, body, keywords, sender address
- Too complex for hand-coded rules
- ✓ Best when patterns are fuzzy and data-driven

1970s and 1990s – the Winter of Machine Learning



Why is ML emerging now?

Global Compute Cost Trends



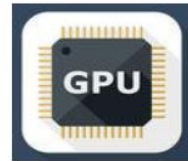
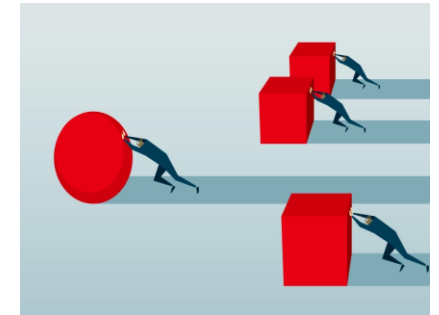
Data “tsunami”



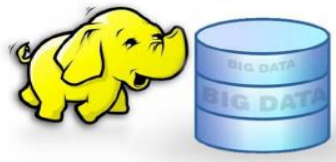
Availability of tools



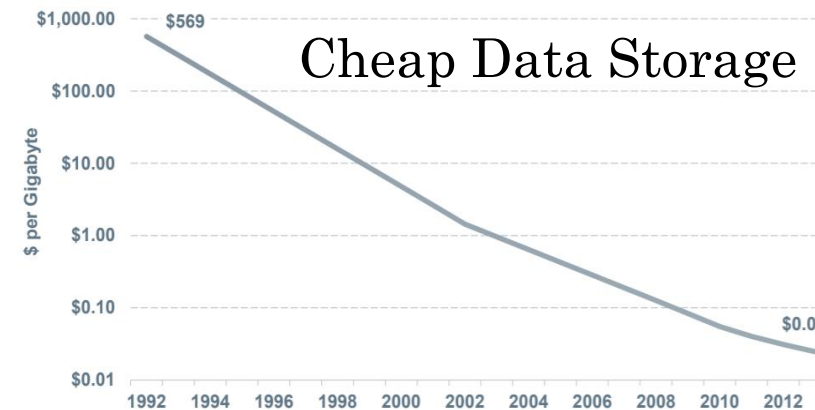
Insatiable desire for efficiency



Convergence



Global Storage Cost Trends

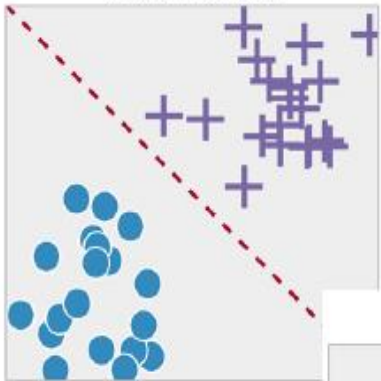


Types of ML?

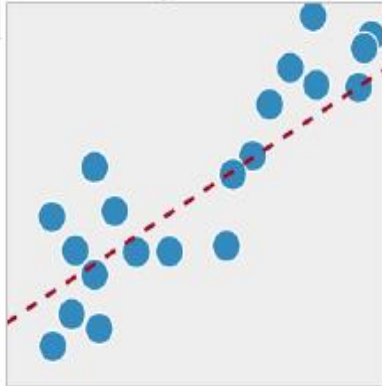
Machine Learning

1. Supervised Learning

Classification

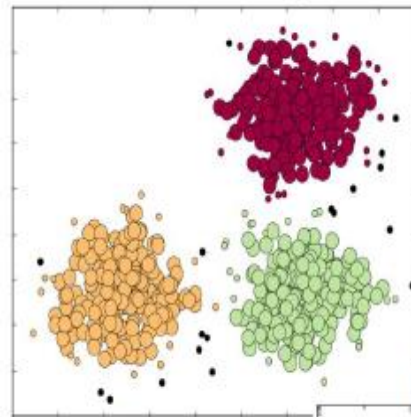


Regression

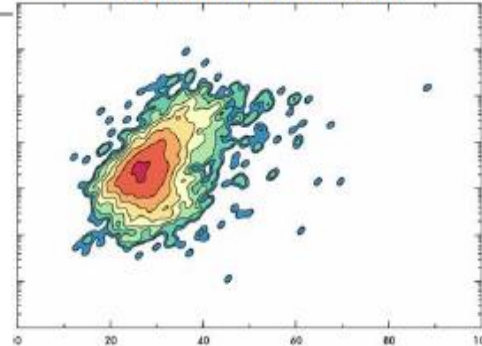


2. Unsupervised Learning

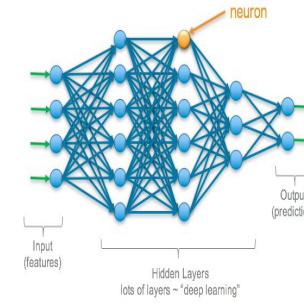
Clustering



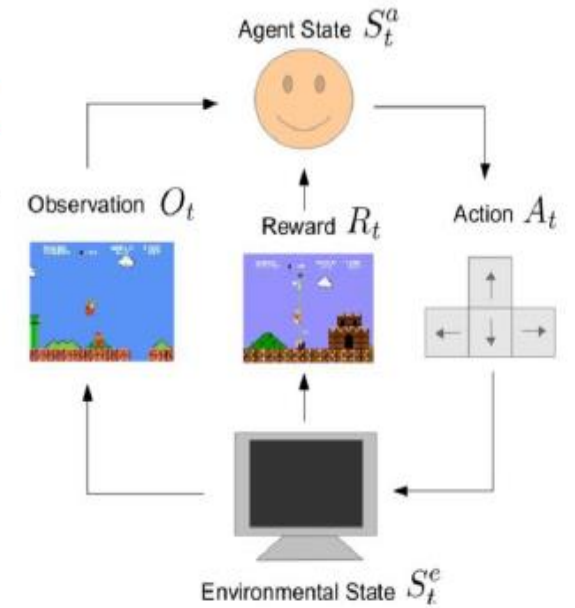
Density Estimation



4. Deep Learning



3. Reinforcement Learning

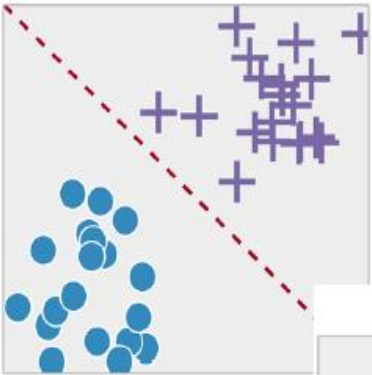


Main Topics We Will Cover

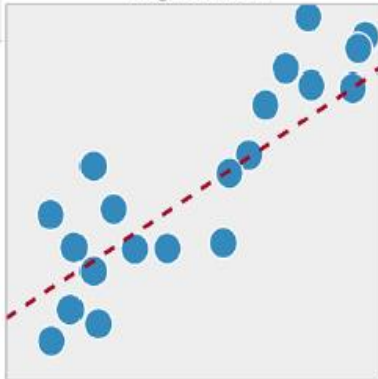
Machine Learning

1. Supervised Learning

Classification

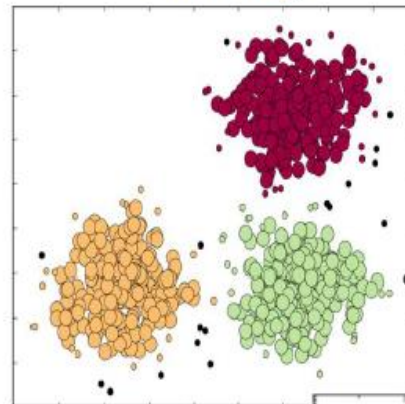


Regression

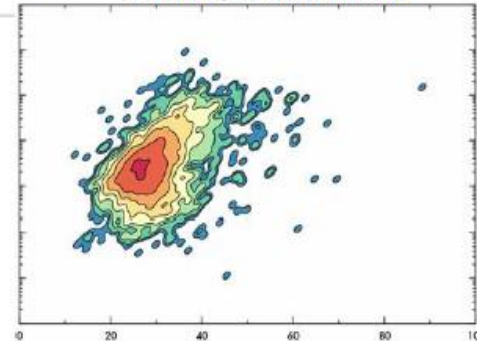


2. Unsupervised Learning

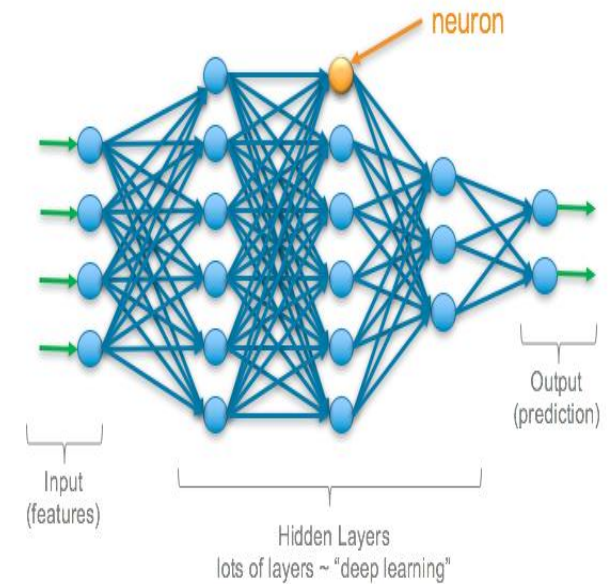
Clustering



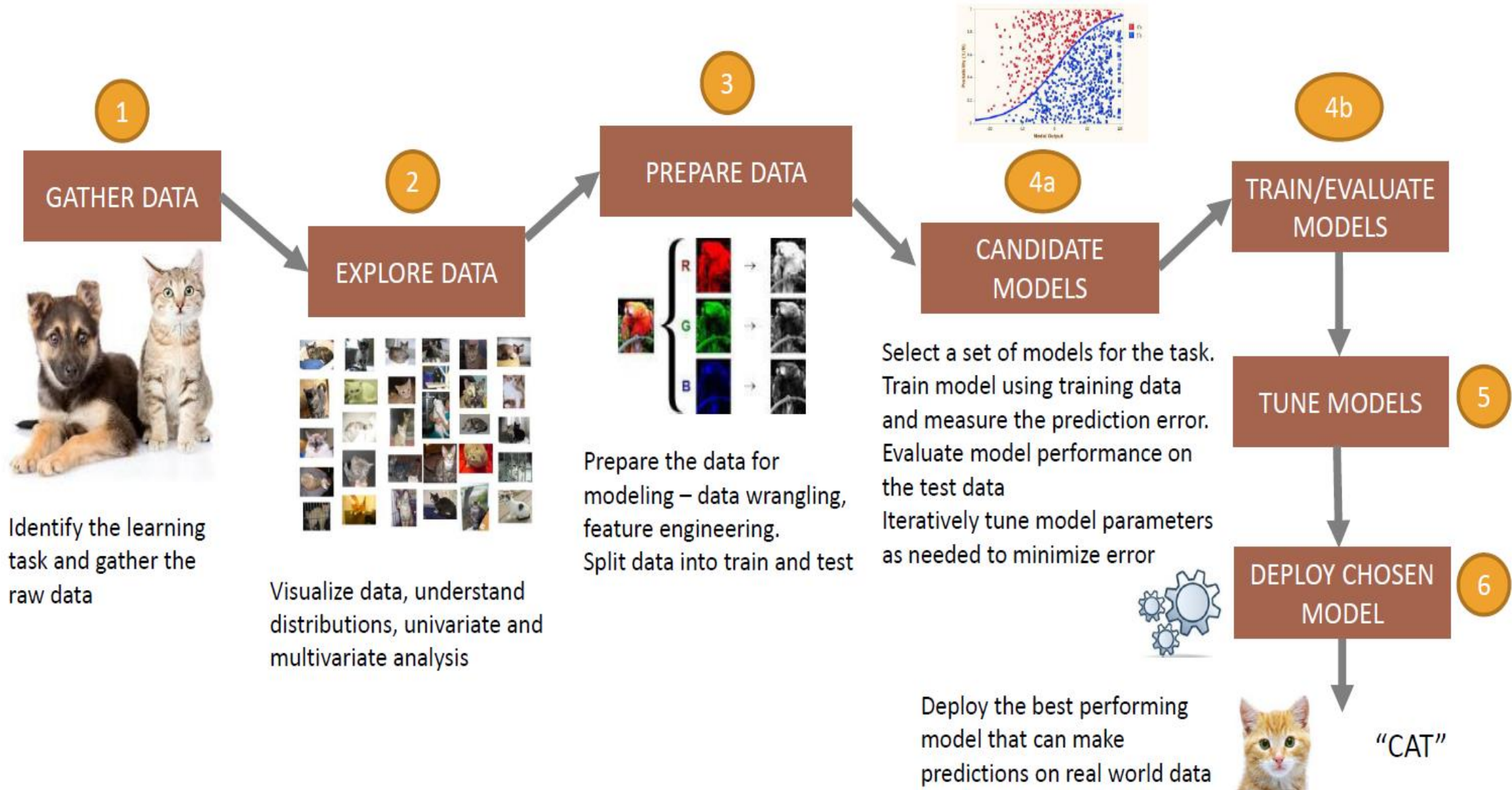
Density Estimation



3. Deep Learning



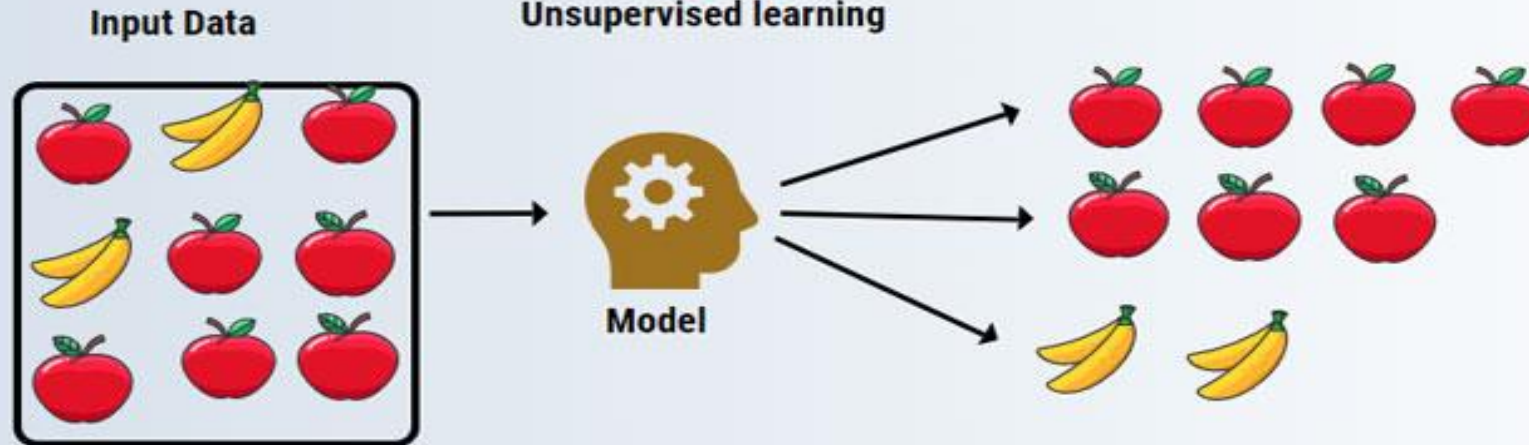
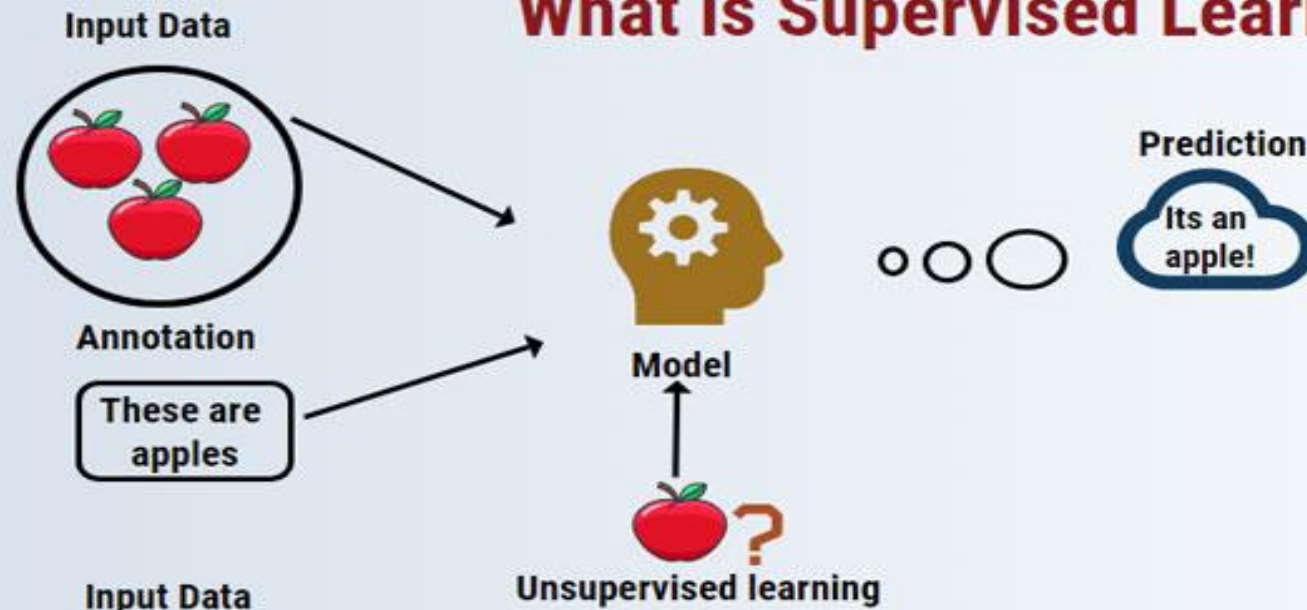
Supervised ML Workflow



	Supervised	Unsupervised
Data	(x, y) where x is data, y is label	x - just data, no labels
Goal	Learn a function to map $x \rightarrow y$ Form a model for $P(x y)$, where y is the label for x	Learn underlying hidden structure of the data Form a model for $P(x)$, where x is an input vector
Methods	Regression, object detection, semantic segmentation, image captioning, etc.	Clustering, dimensionality reduction, feature learning, density estimation, etc.
Example	  "Cat"	  

Supervised vs Unsupervised ML

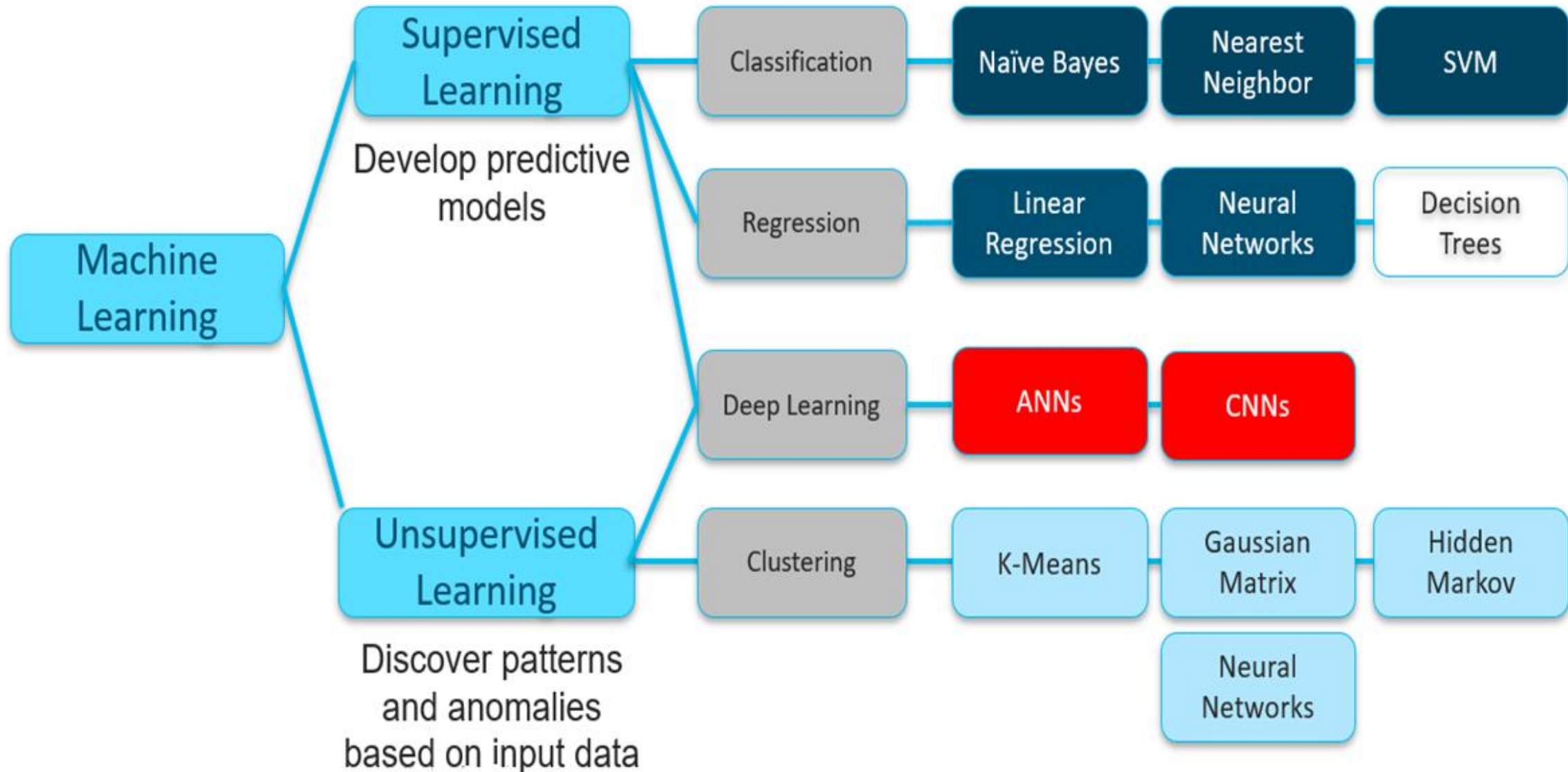
What is Supervised Learning?



www.educba.com

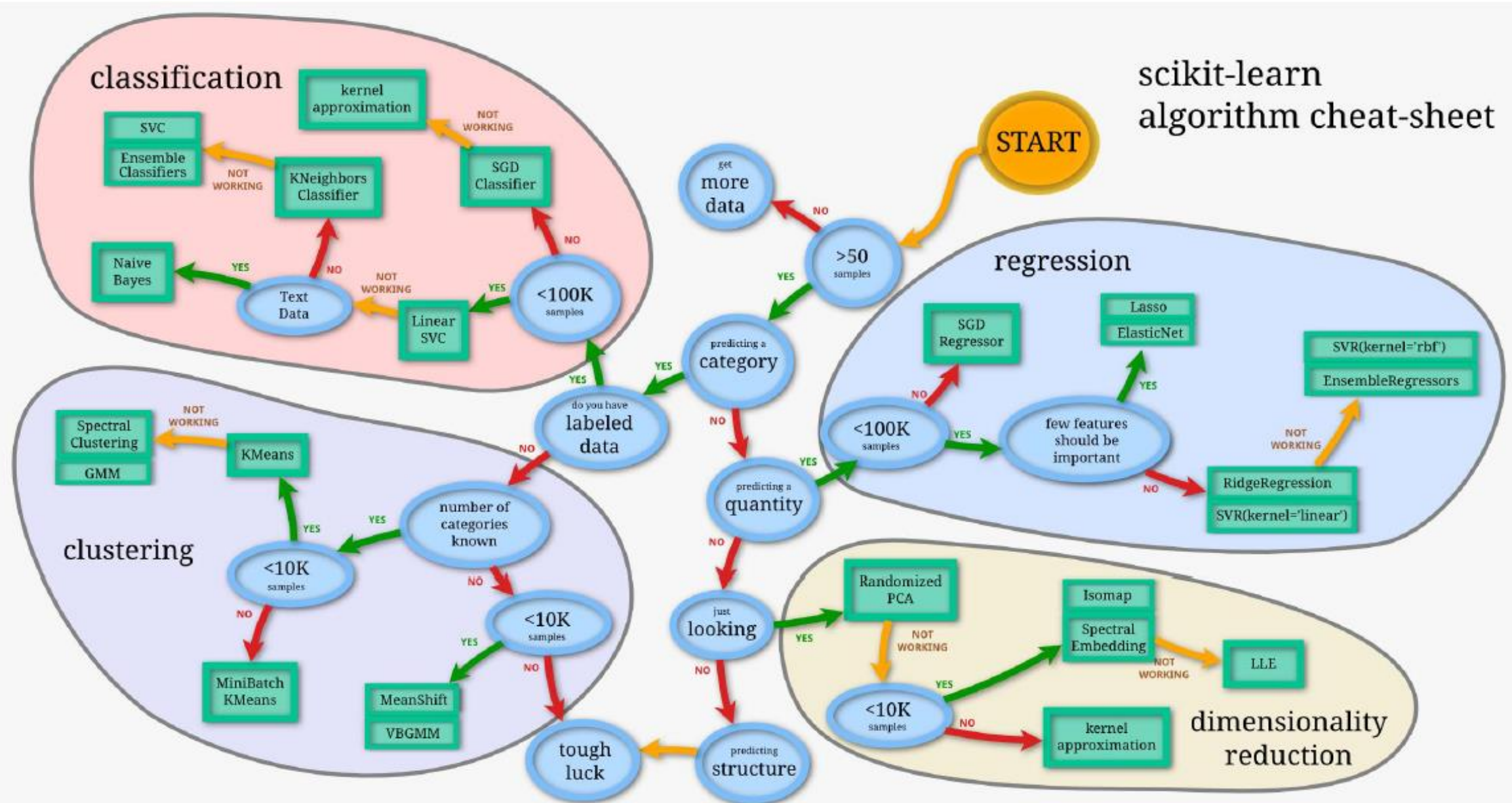
Supervised vs Unsupervised ML

Different ML Algorithms



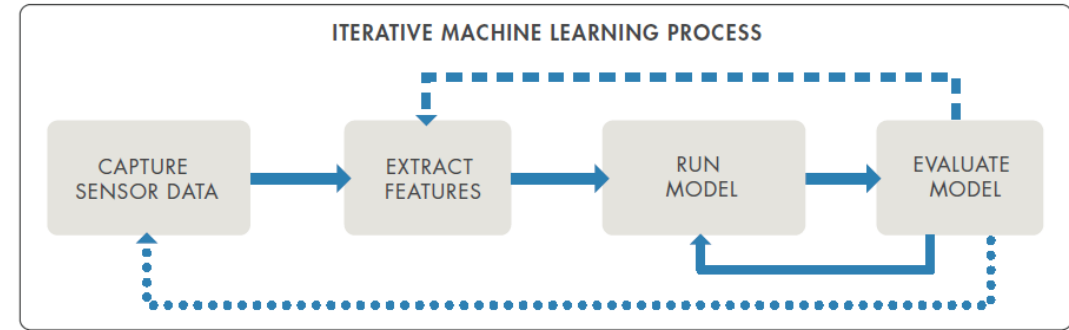
How to Choose a ML Algorithm?

scikit-learn
algorithm cheat-sheet

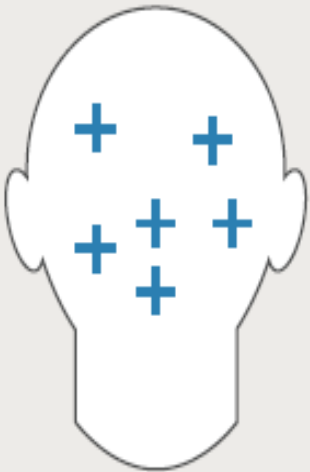


When Should You Use Machine Learning?

Consider using machine learning when you have a complex task or problem involving a large amount of data and lots of variables, but no existing formula or equation. For example, machine learning is a good option if you need to handle situations like these:



Hand-written rules and equations are too complex—as in face recognition and speech recognition.



[Image Source](#)

The rules of a task are constantly changing—as in fraud detection from transaction records.



The nature of the data keeps changing, and the program needs to adapt—as in automated trading, energy demand forecasting, and predicting shopping trends.

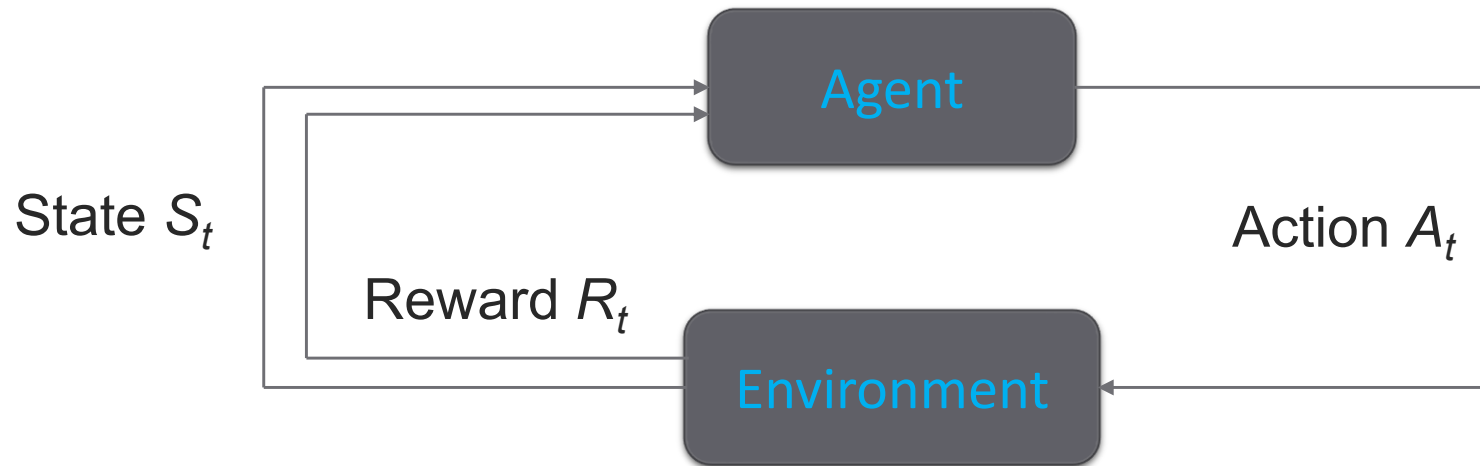
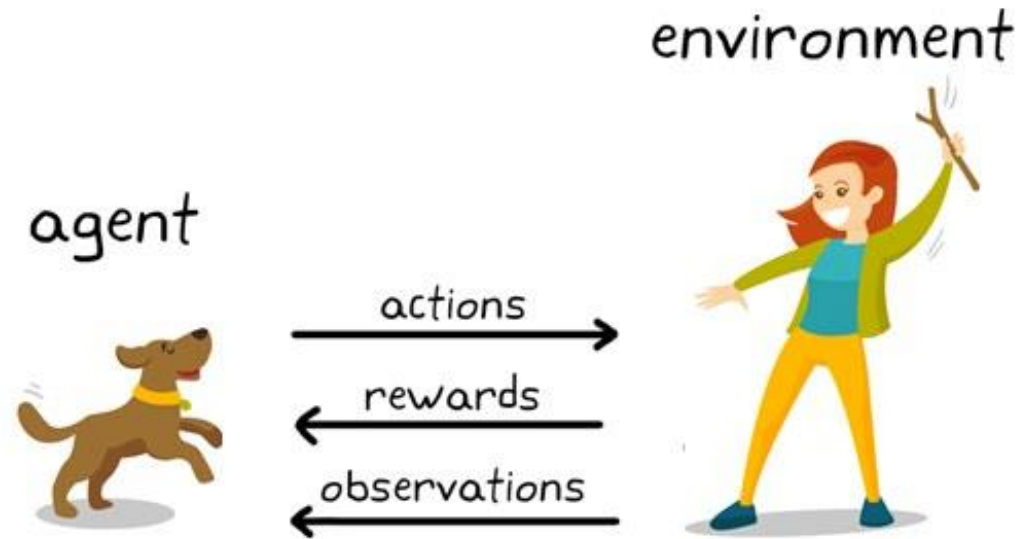


What Type of Questions can ML Answer?

Questions	Description	Examples
Is this A or B?	Classification: Questions that have two or more possible answers.	Which animal is in this image? Will this customer click on the top link?
Is this an anomaly ?	Anomaly Detection: Questions about events that seems out of the ordinary.	Is this pressure reading unusual? Is this combination of purchases normal for this customer ?
How much/how many ?	Forecasting/Regression: When we are looking for a number	What will the temperature be next Tuesday? How many customers will we acquire next quarter ?
How is it organized ?	Segmentation/Clustering: Knowing if there a hidden structure in our data points.	Which shoppers have similar tastes in produce? Which viewers like the same kind of movies?
What should I do next?	Recommendation: Deciding the optimal future actions based on past information	Where should I place this ad on the webpage so that the viewer is most likely to click it ? How many shares of this stock should I buy ?

Reinforcement Learning

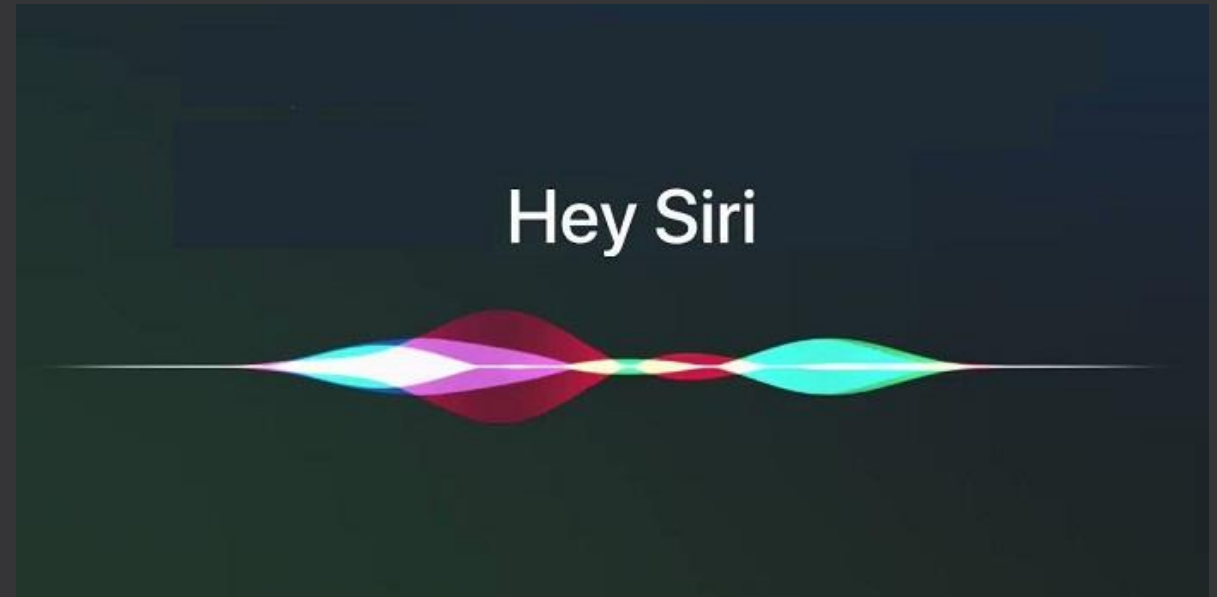
- A semi-supervised learning model.
- No training data or correct/incorrect guidance needs to be given.
- Involves behavioural psychology.
- In summary, a lot of trial and error.



What is Deep Learning?

Questions worth wondering about?

- How does an autonomus car drive?

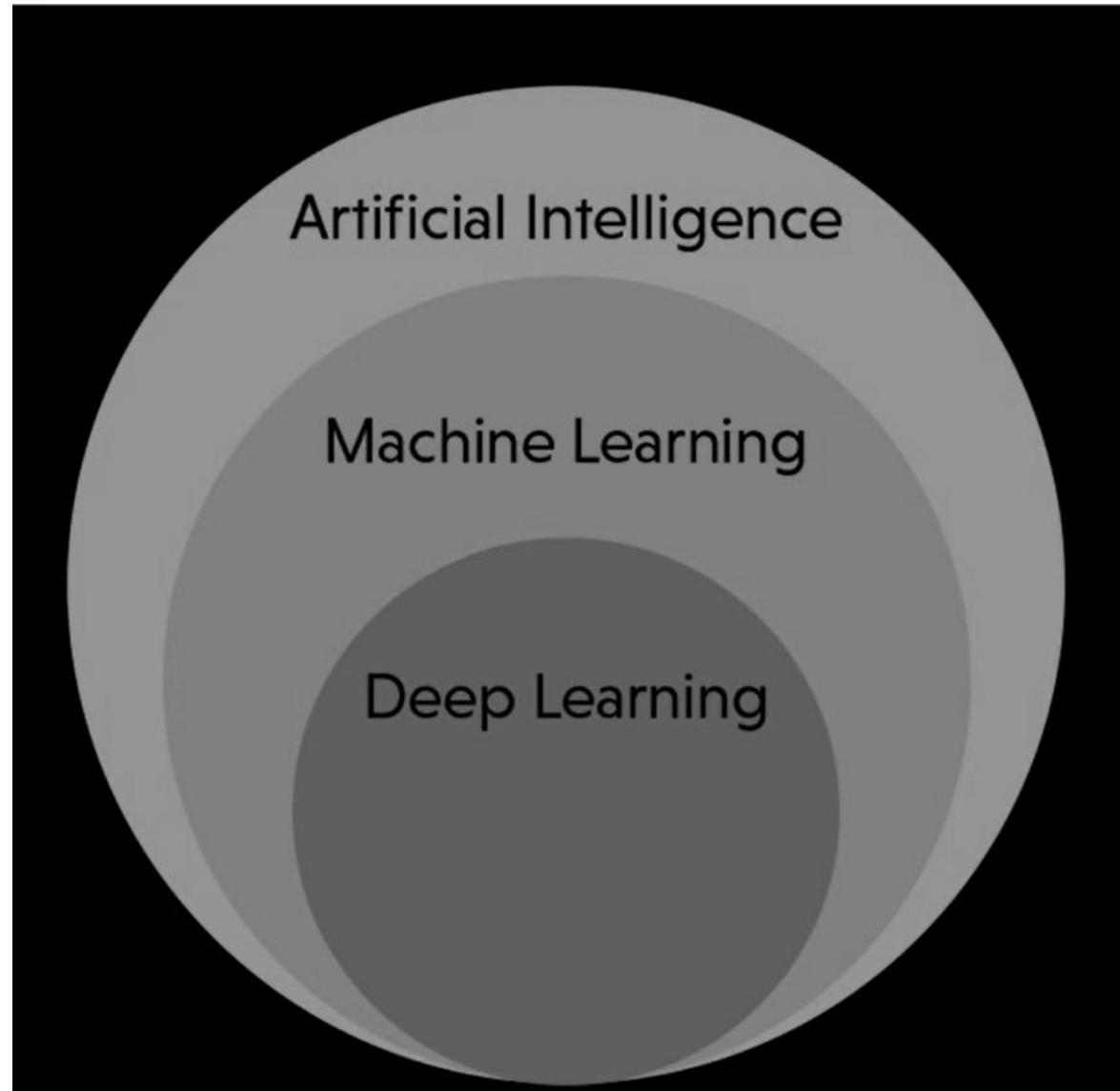


- How does Siri respond to your voice commands?
- It's not MAGIC, it's DEEP LEARNING

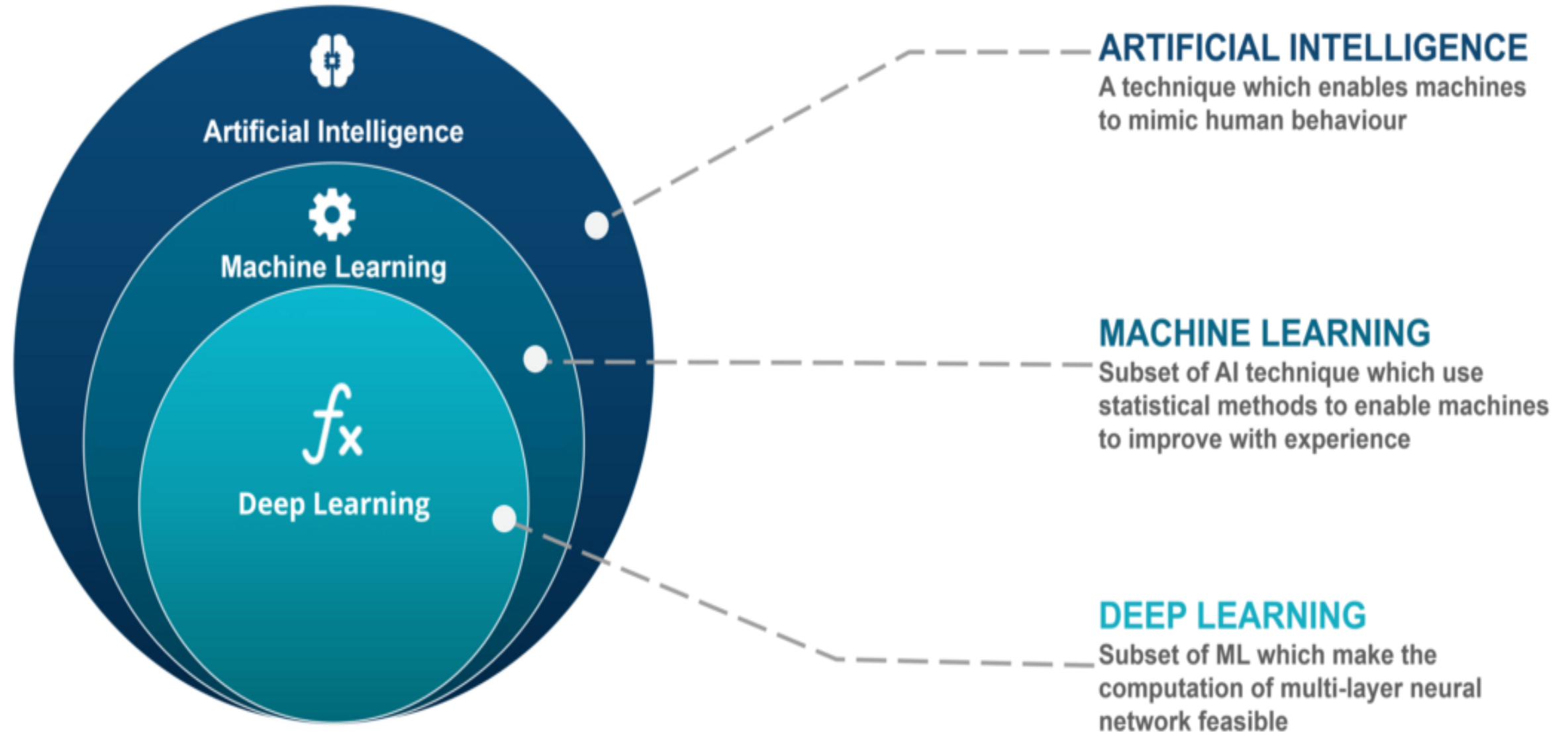
What is Deep Learning?

Did you know:

- Deep Learning is a sub class of Machine Learning.
- Uses Artificial Neural Networks (ANN) to learn from data.
- Artificial Neuron is inspired from the structure of human brain neuron.

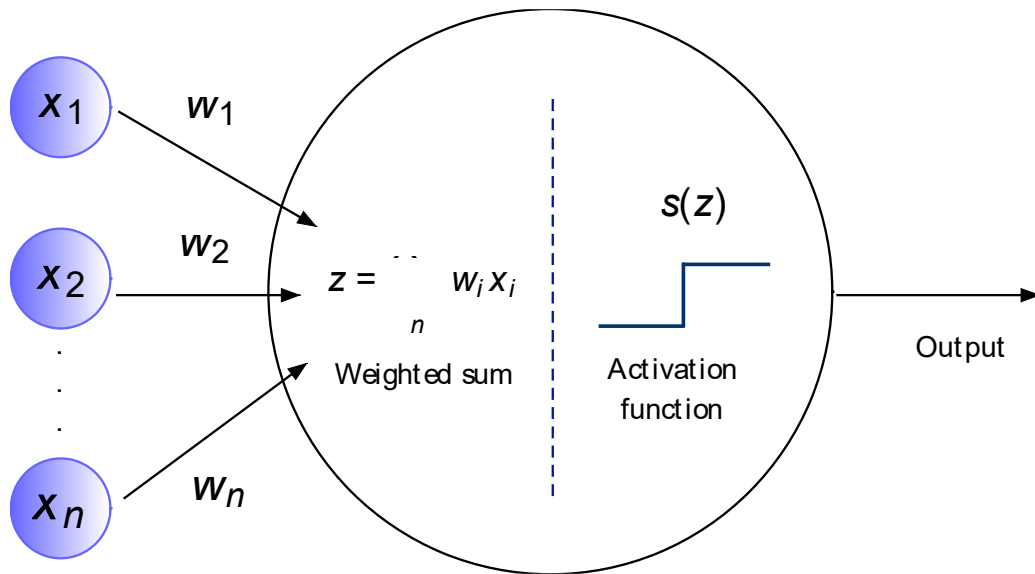
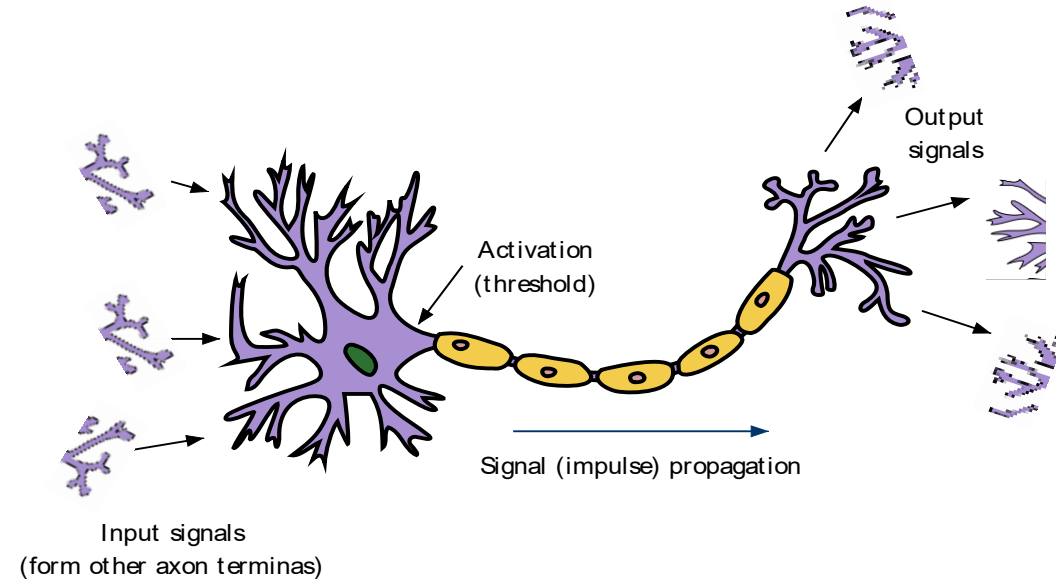


AI & ML & DL



Neural Networks: Overview

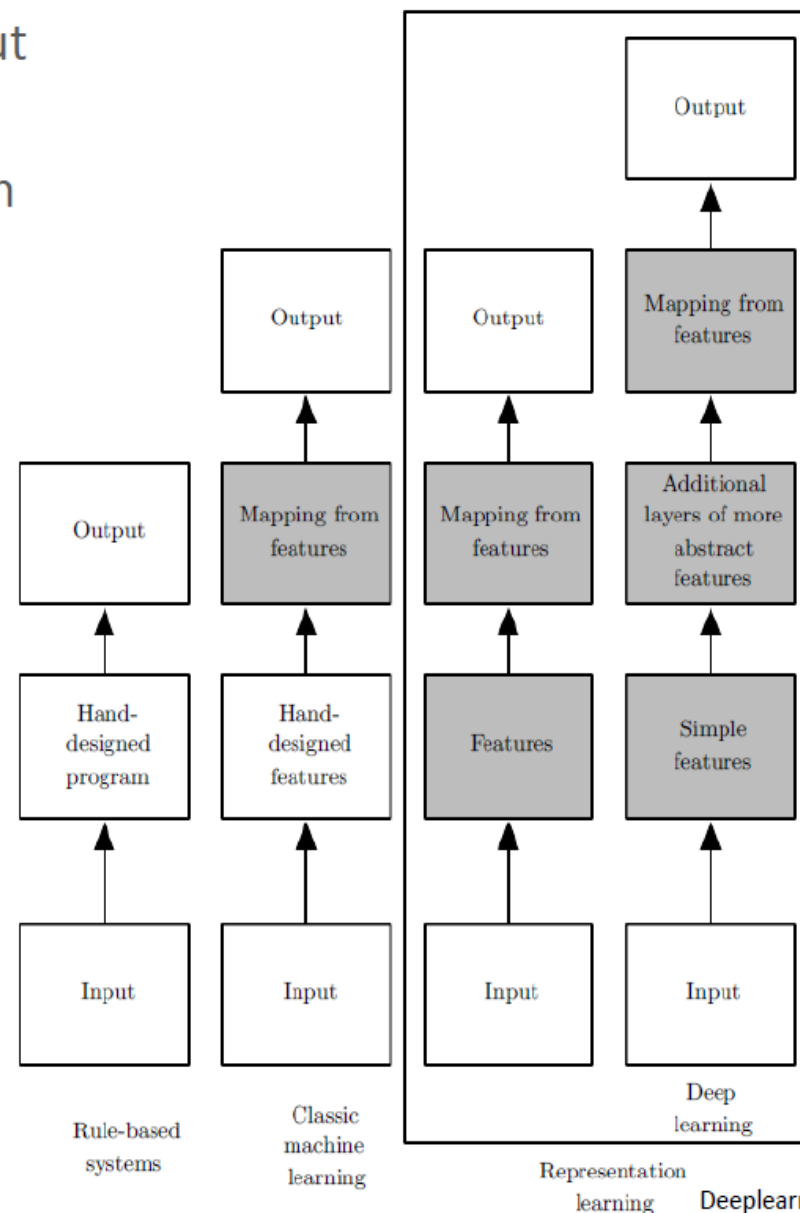
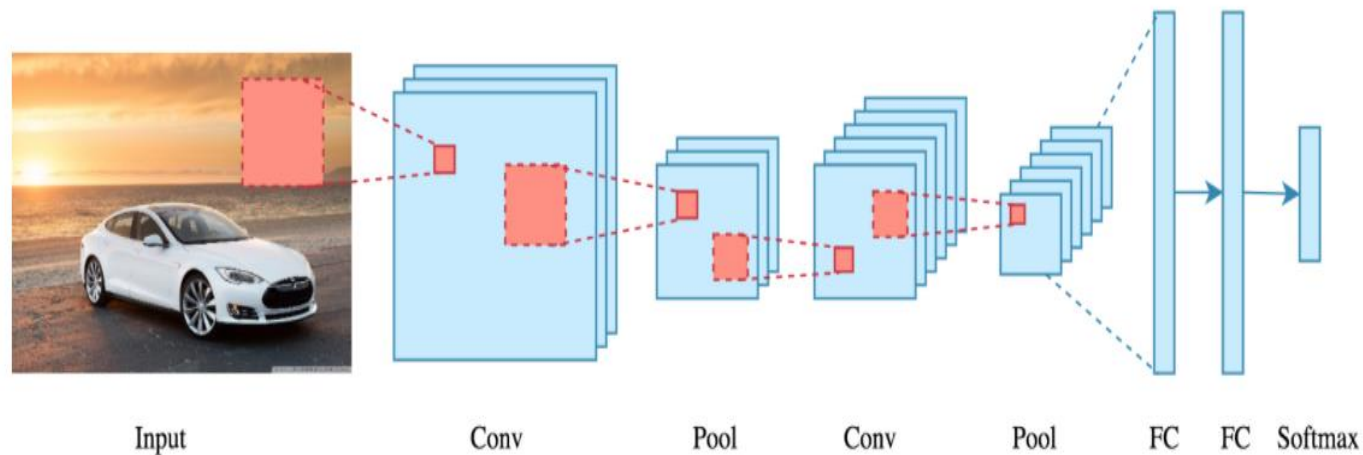
- Origins: Algorithms that try to mimic the brain
- Were very widely used in the 1980s and 90s: popularity diminished in the late 1990s.
- Recent resurgence: State-of-the-art technique for many applications.



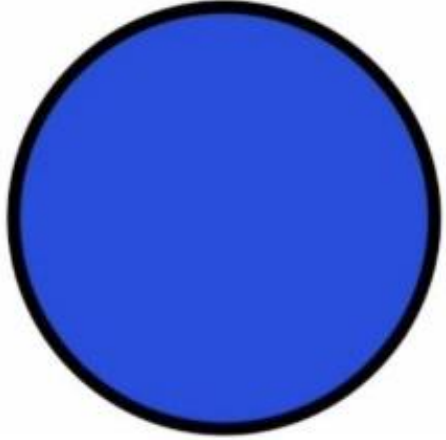
1. Inputs arrive from other neurons / nodes.
2. Receiving neuron / node processes the inputs $\rightarrow$ weighted sum.
3. Receiving neuron / node applies an activation function (sigmoid).
4. Output of activation function passed to subsequent neurons / nodes.

Understanding Deep Learning

- Deep Learning allows computers to build complex concepts out of simple concepts
- By feeding sufficient data into deep neural networks, it is often possible to learn better representations than handcrafted features



Understanding Deep Learning

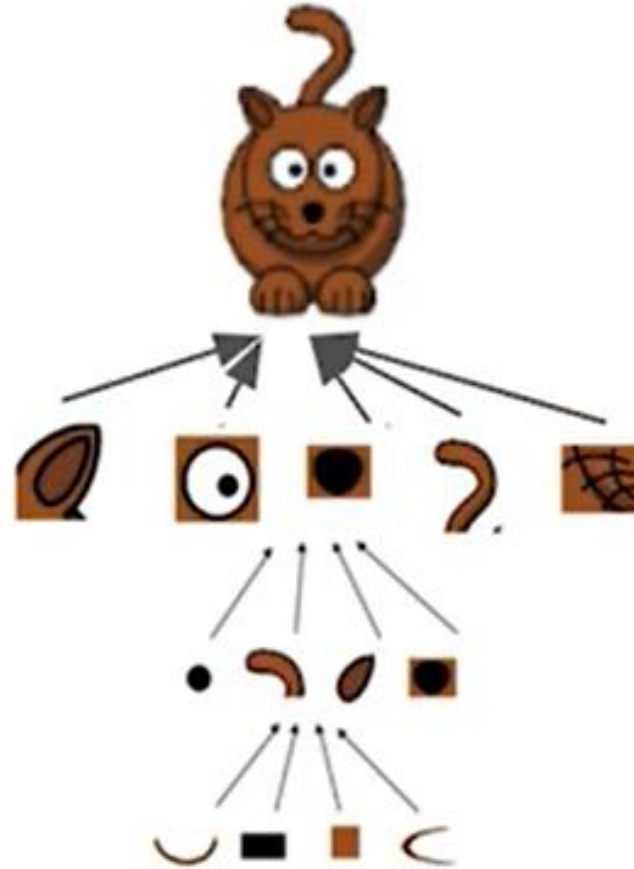


1. Four Lines
2. They are connected, closed, perpendicular to each other
3. All lines are equal in length

REMEMBER THIS:

A Deep Learning Model breaks a complex concept into a hierarchy of simpler concepts.

Understanding Deep Learning



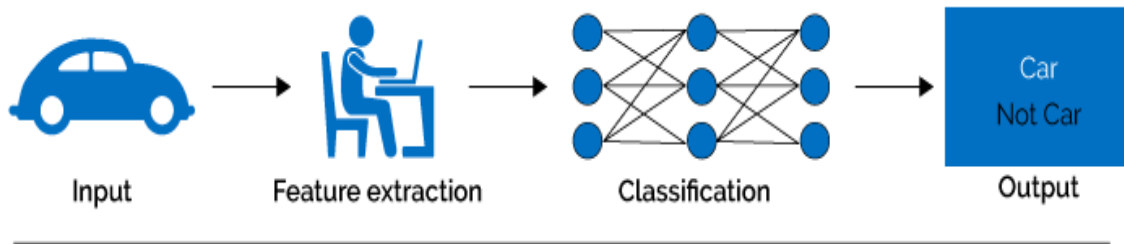
1. Identify edges.
2. Builds on this hierarchy to find combination of shapes and edges.
3. Identifies which of these features are responsible for Cat vs. Dog classification.

Deep Learning vs. Machine Learning

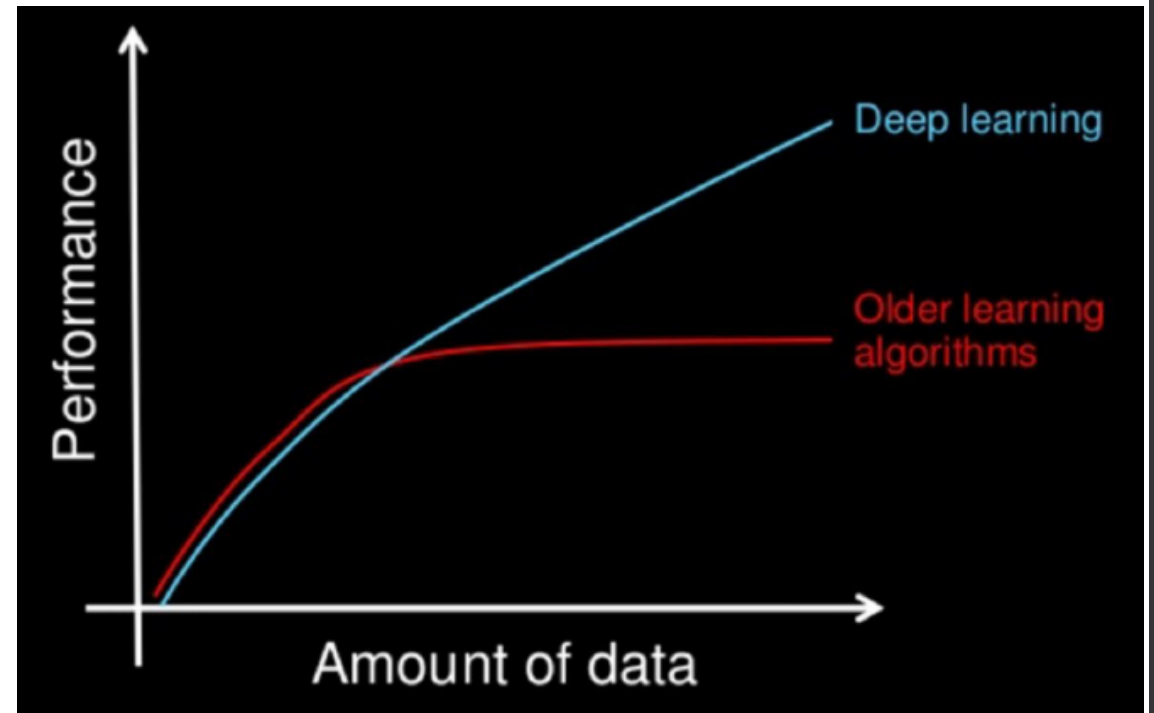
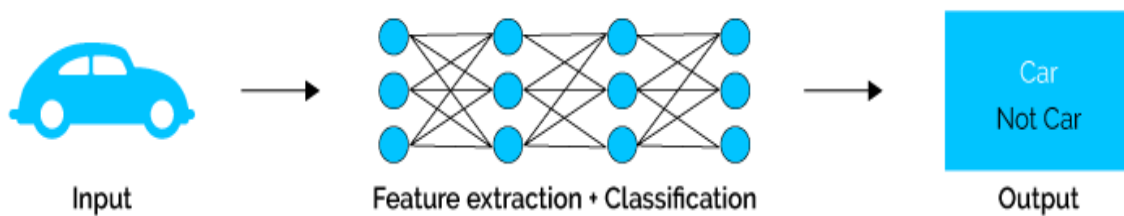
Automatic
Feature
Extraction

Performs well on
More Data

Machine Learning



Deep Learning



Deep Learning vs. Machine Learning

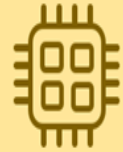
Requires More
Computation
Power

Less
Interpretable



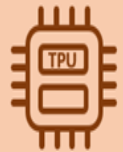
CPU

- Small models
- Small datasets
- Useful for design space exploration



GPU

- Medium-to-large models, datasets
- Image, video processing
- Application on CUDA or OpenCL



TPU

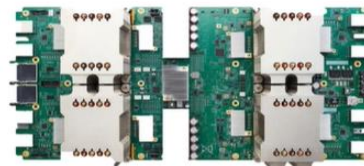
- Matrix computations
- Dense vector processing
- No custom TensorFlow operations



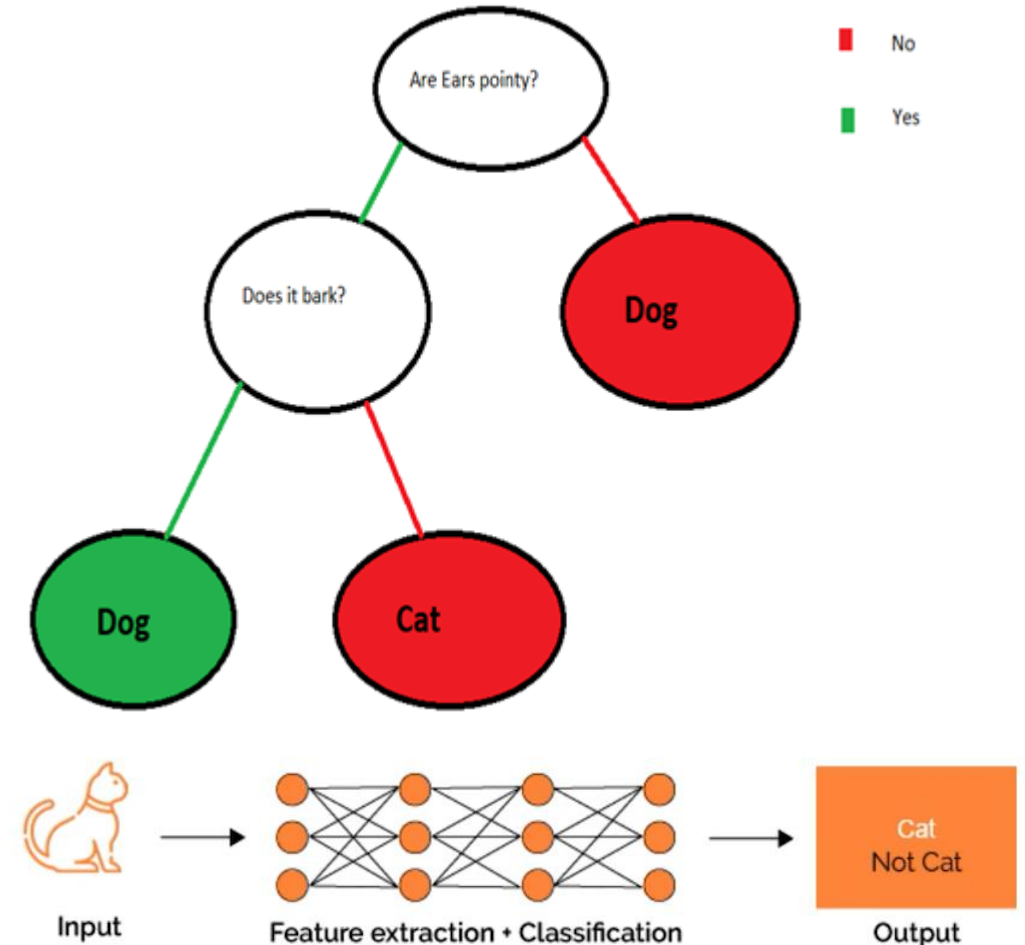
CPU



GPU

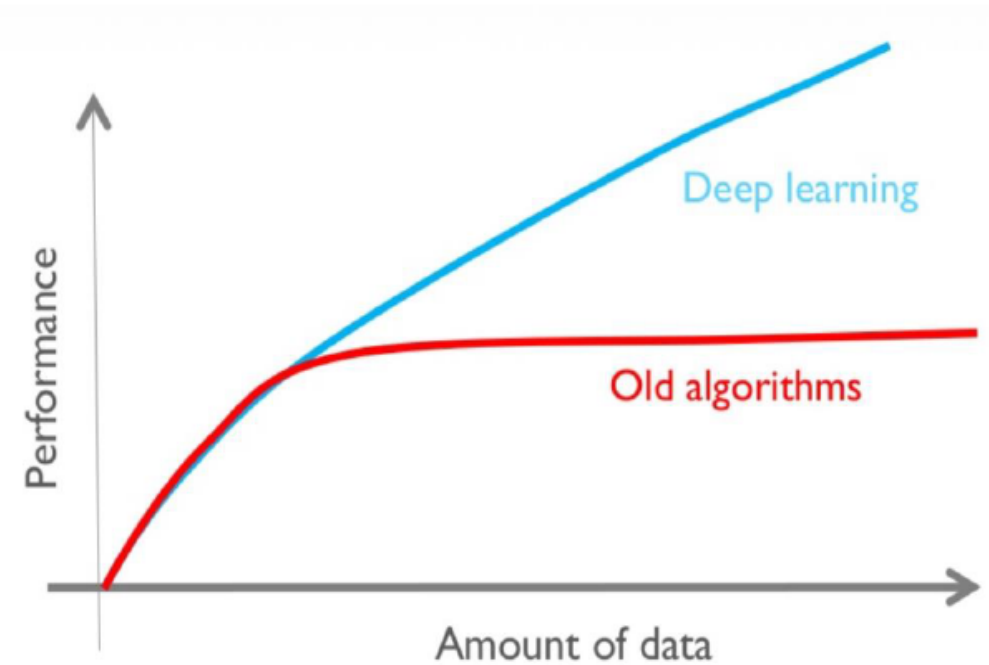
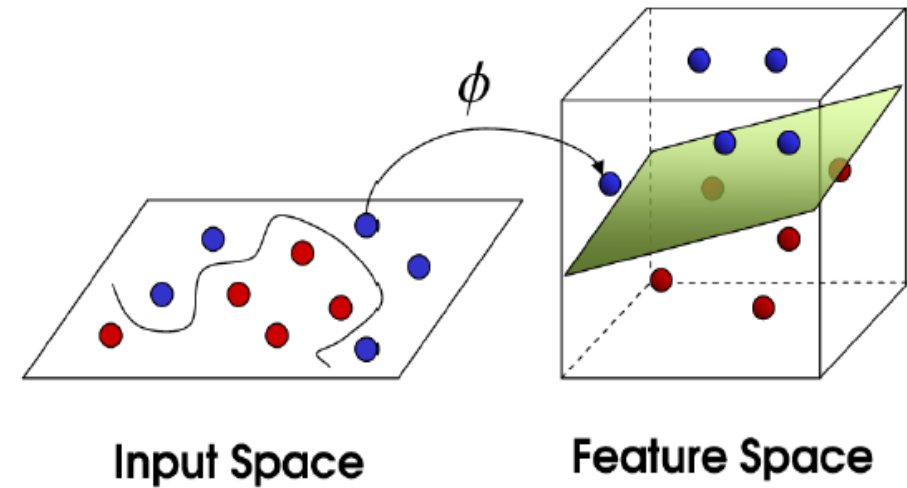


TPU



Why Deep Learning?

- ML Feature engineering
 - Handcrafting features is time-consuming
 - Features are usually over-specified or incomplete
 - Effort is repetitive for each task/domain
- Higher performance than ML
 - Deep learning yields higher performance as training data grows
- Can be applied to complex tasks
 - Computer Vision, NLP

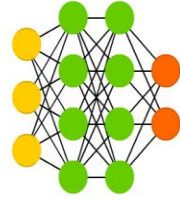


Neural Networks

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-  Backfed Input Cell
-  Input Cell
-  Noisy Input Cell
-  Hidden Cell
-  Probabilistic Hidden Cell
-  Spiking Hidden Cell
-  Output Cell
-  Match Input Output Cell
-  Recurrent Cell
-  Memory Cell
-  Different Memory Cell
-  Kernel
-  Convolution or Pool

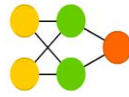
Deep Feed Forward (DFF)



Perceptron (P)



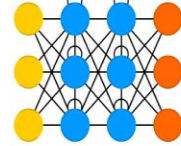
Feed Forward (FF)



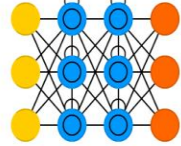
Radial Basis Network (RBF)



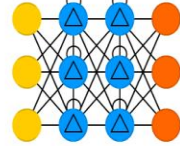
Recurrent Neural Network (RNN)



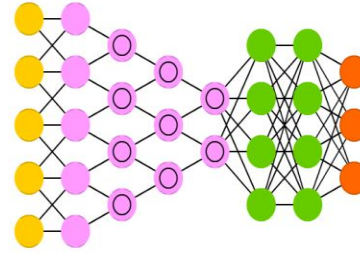
Long / Short Term Memory (LSTM)



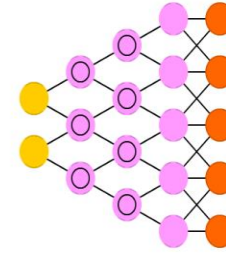
Gated Recurrent Unit (GRU)



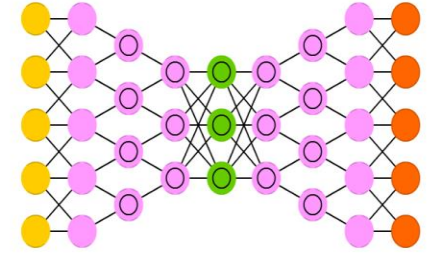
Deep Convolutional Network (DCN)



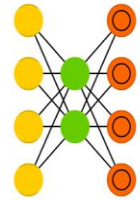
Deconvolutional Network (DN)



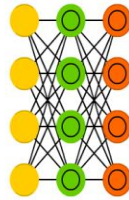
Deep Convolutional Inverse Graphics Network (DCIGN)



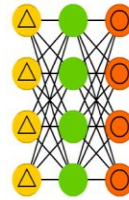
Auto Encoder (AE)



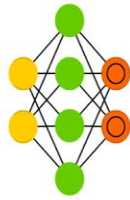
Variational AE (VAE)



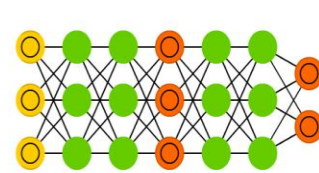
Denoising AE (DAE)



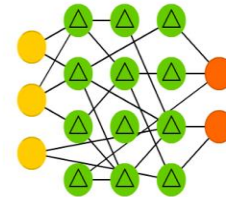
Sparse AE (SAE)



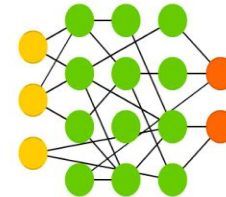
Generative Adversarial Network (GAN)



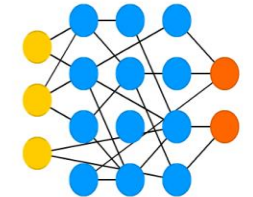
Liquid State Machine (LSM)



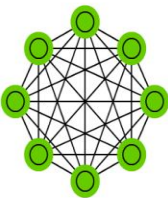
Extreme Learning Machine (ELM)



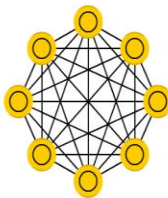
Echo State Network (ESN)



Markov Chain (MC)



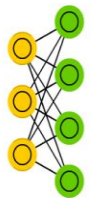
Hopfield Network (HN)



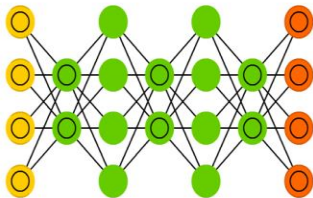
Boltzmann Machine (BM)



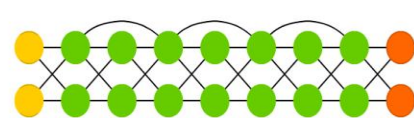
Restricted BM (RBM)



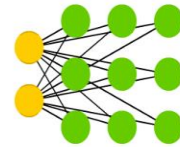
Deep Belief Network (DBN)



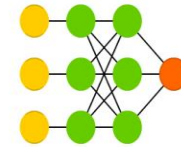
Deep Residual Network (DRN)



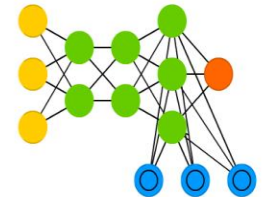
Kohonen Network (KN)



Support Vector Machine (SVM)



Neural Turing Machine (NTM)

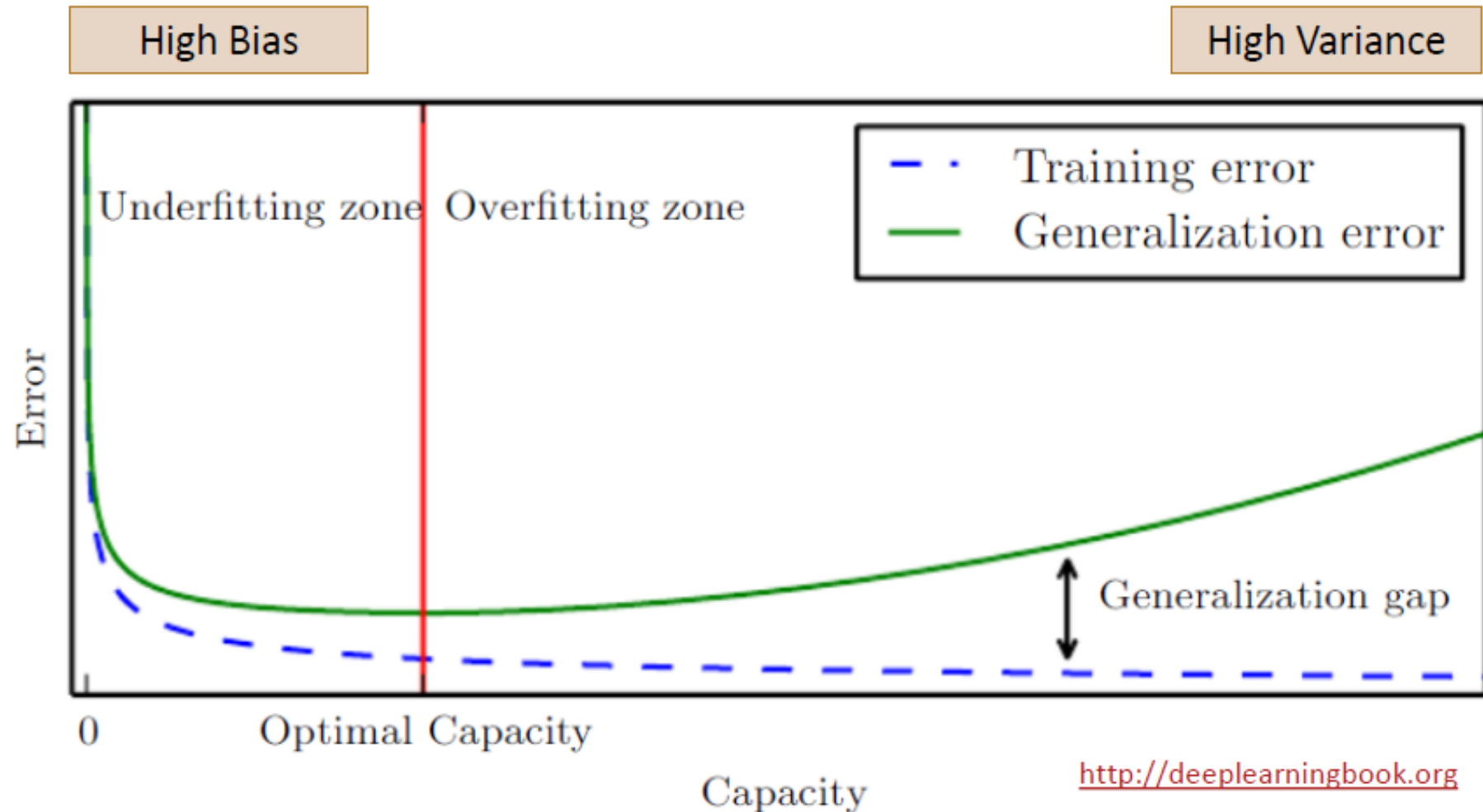


Neural Network Types

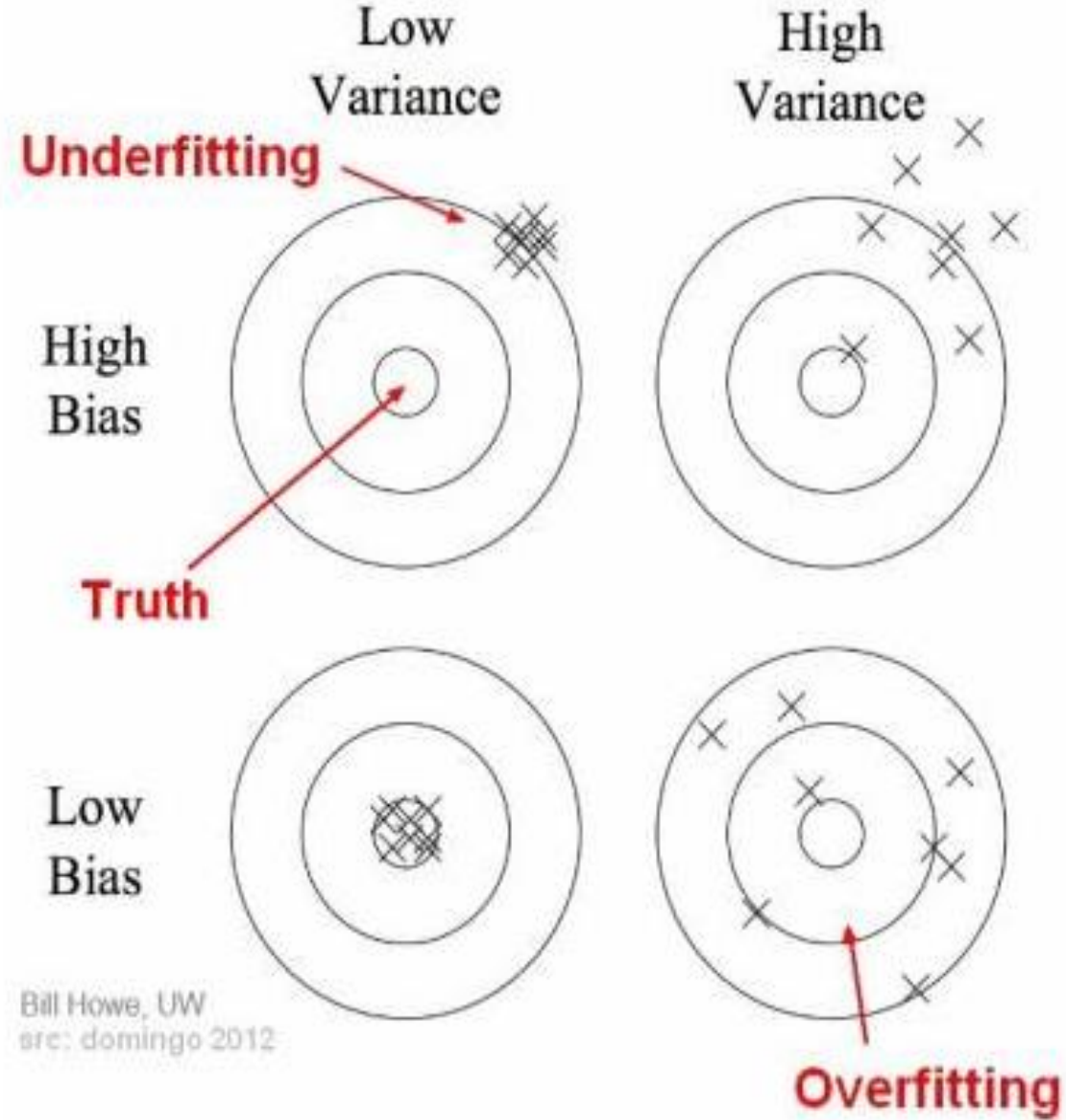
ML Challenges

Generalization Error

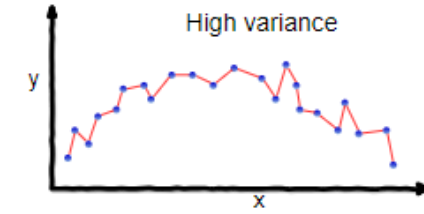
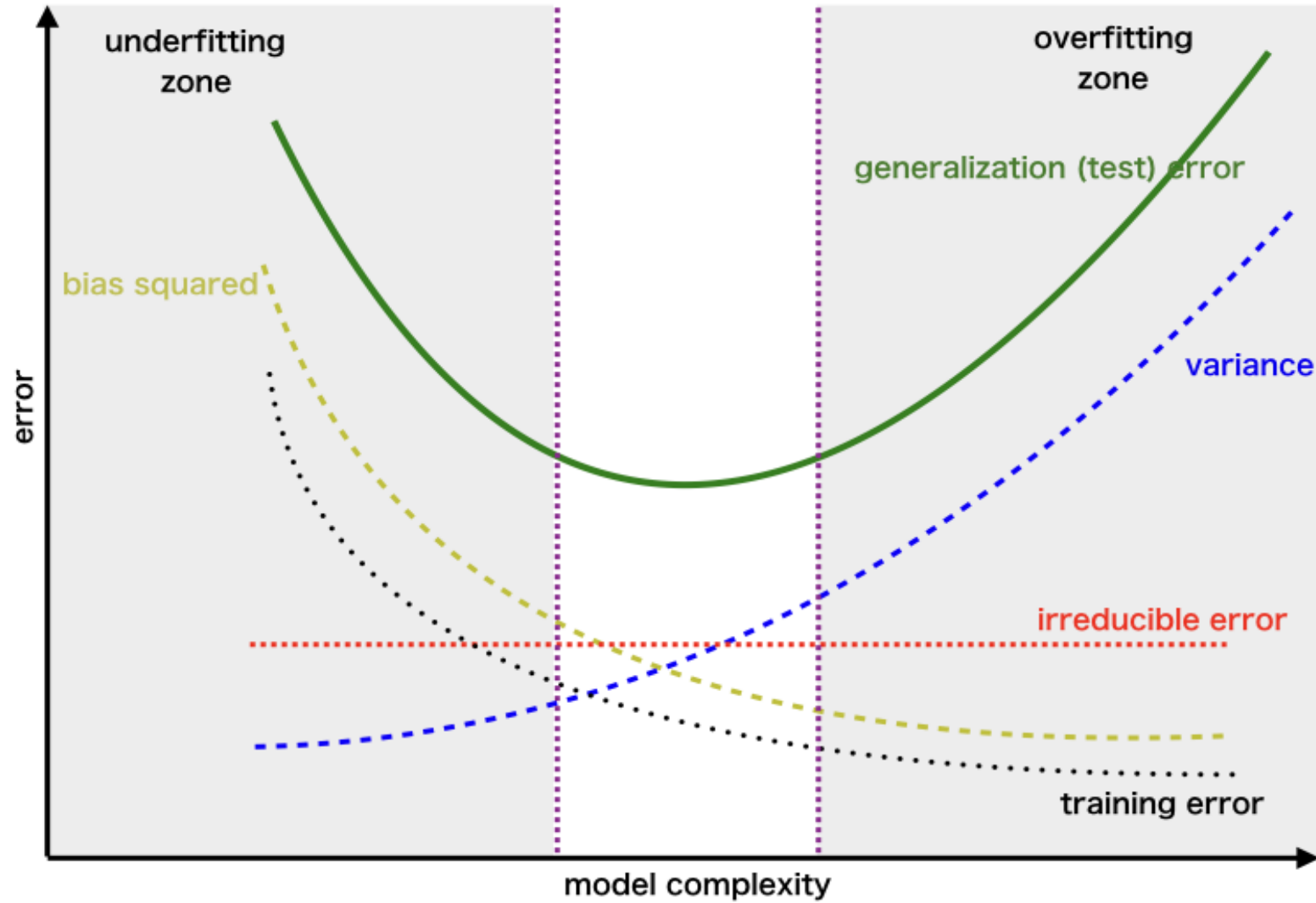
- The central challenge in ML is to perform well on new, previously unseen inputs



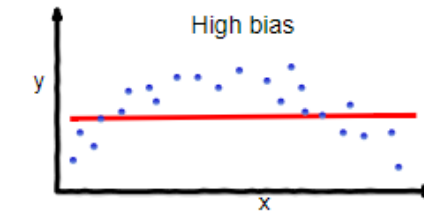
Bias-Variance Tradeoff



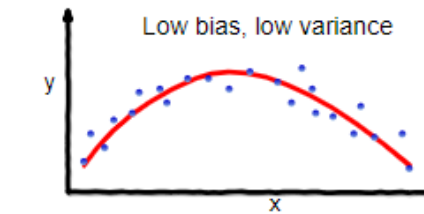
Bias-Variance Tradeoff



overfitting

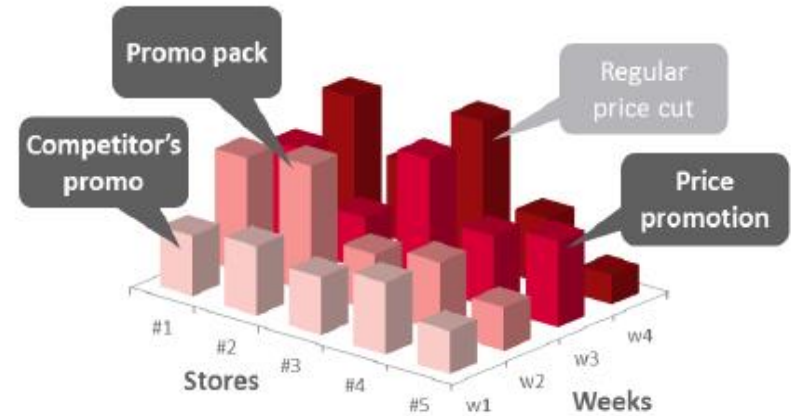
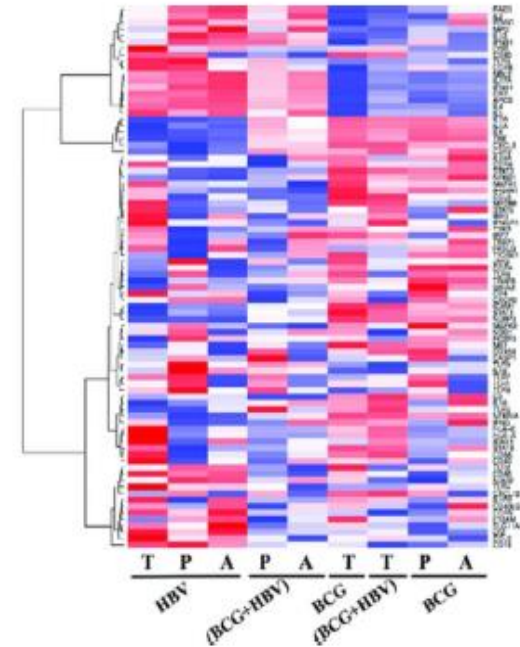
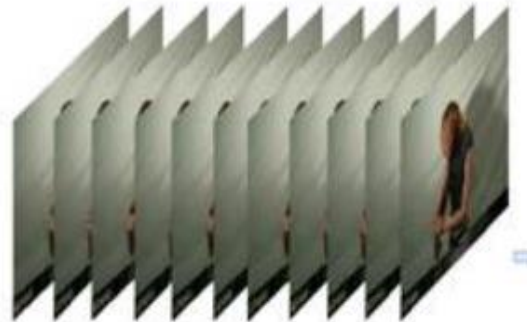
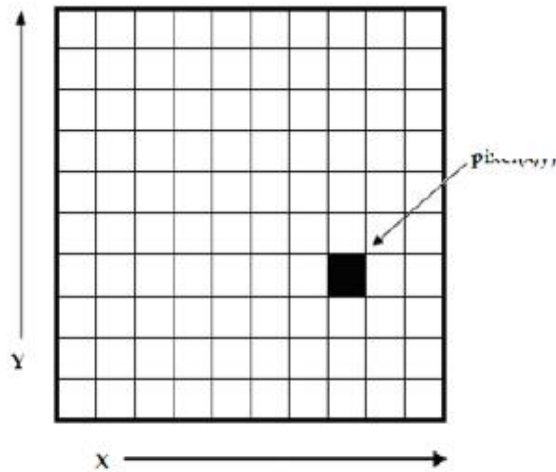
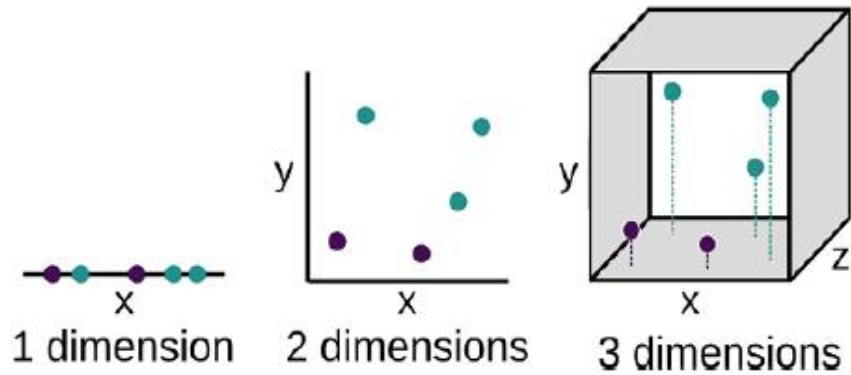


underfitting

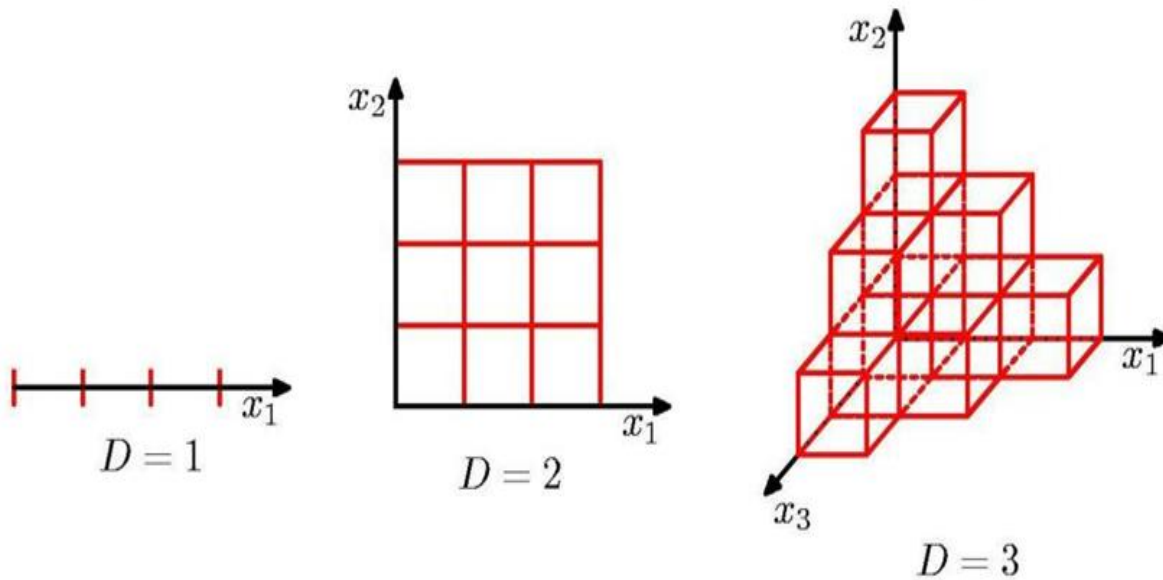


Good balance

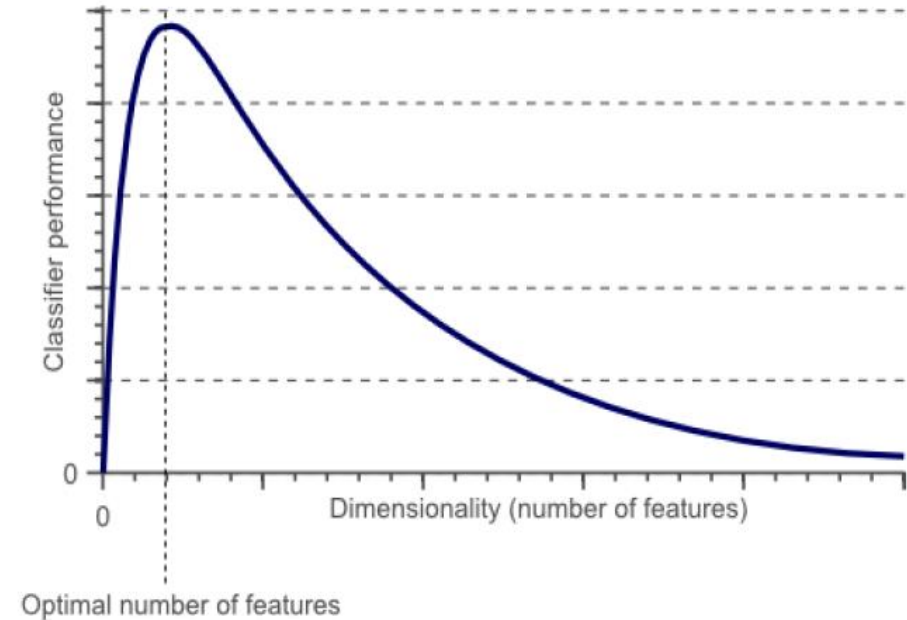
High Dimensional Data



Curse of Dimensionality



- ▶ No. of cells grow exponentially with D
- ▶ Need exponentially large no. of training data points
- ▶ Not a good approach for more than a few dimensions!

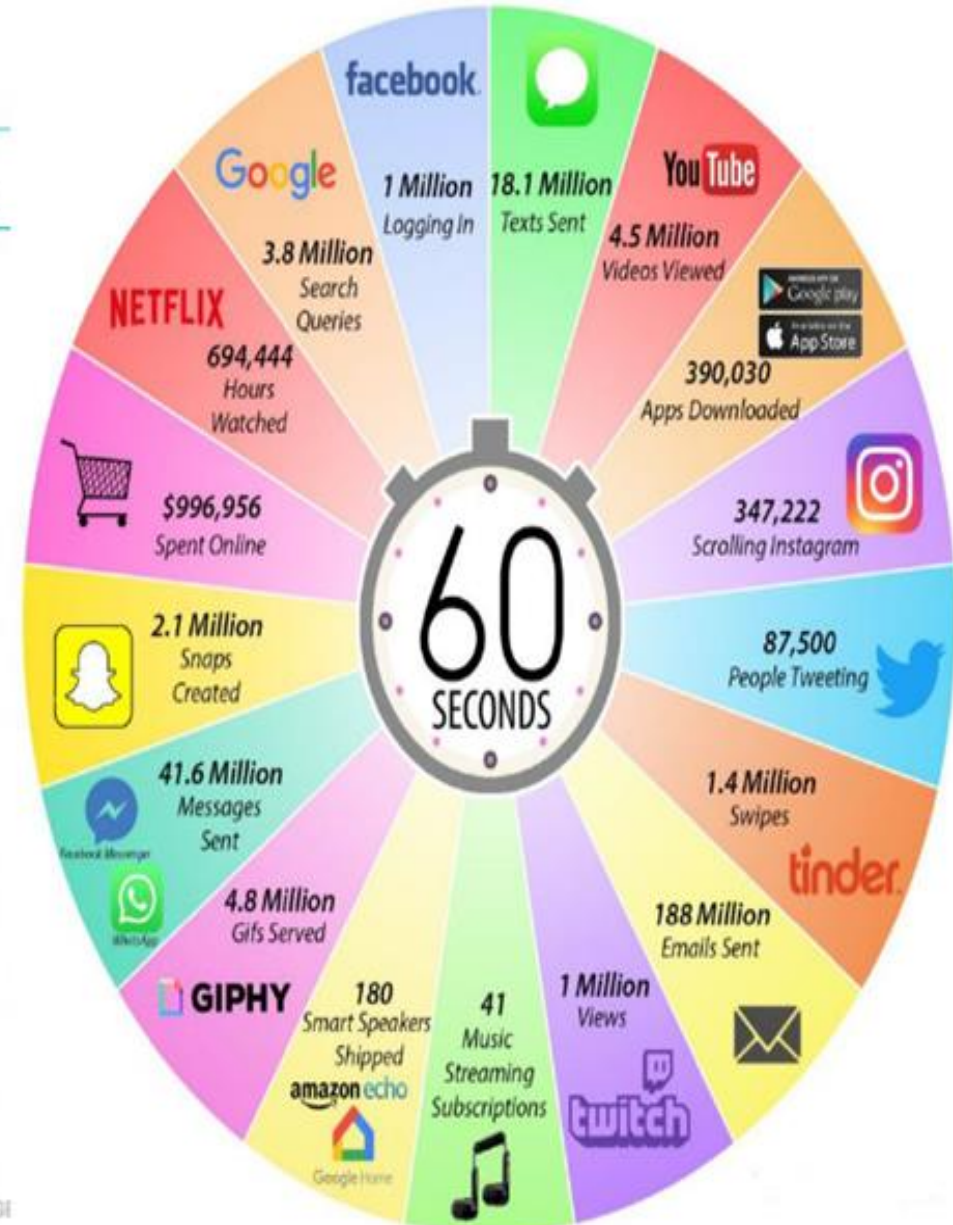
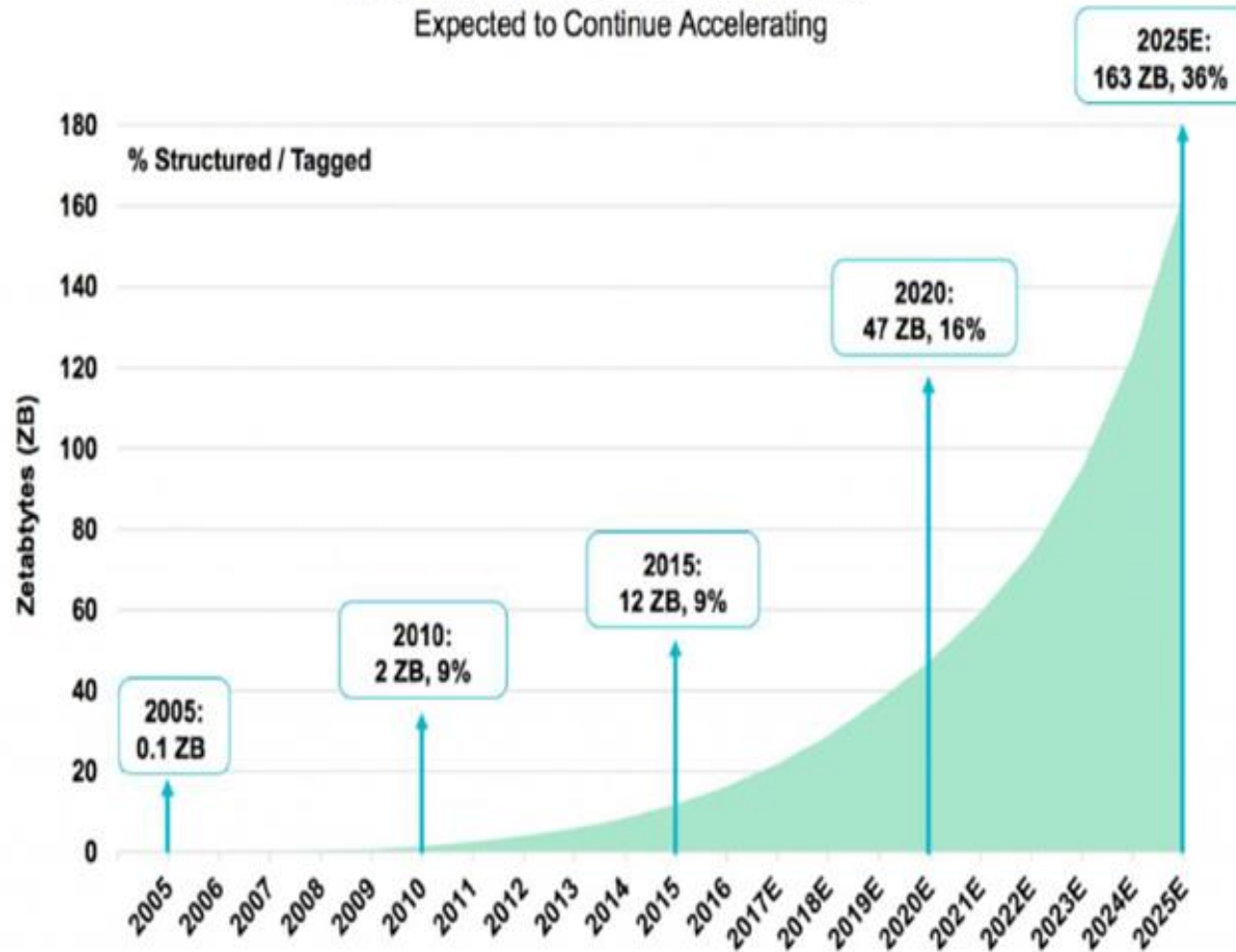


Reference: Christopher M Bishop: Pattern Recognition & Machine Learning, 2006 Springer

- As the number of features (dimensions) grows, the amount of data we need to generalize grows exponentially.

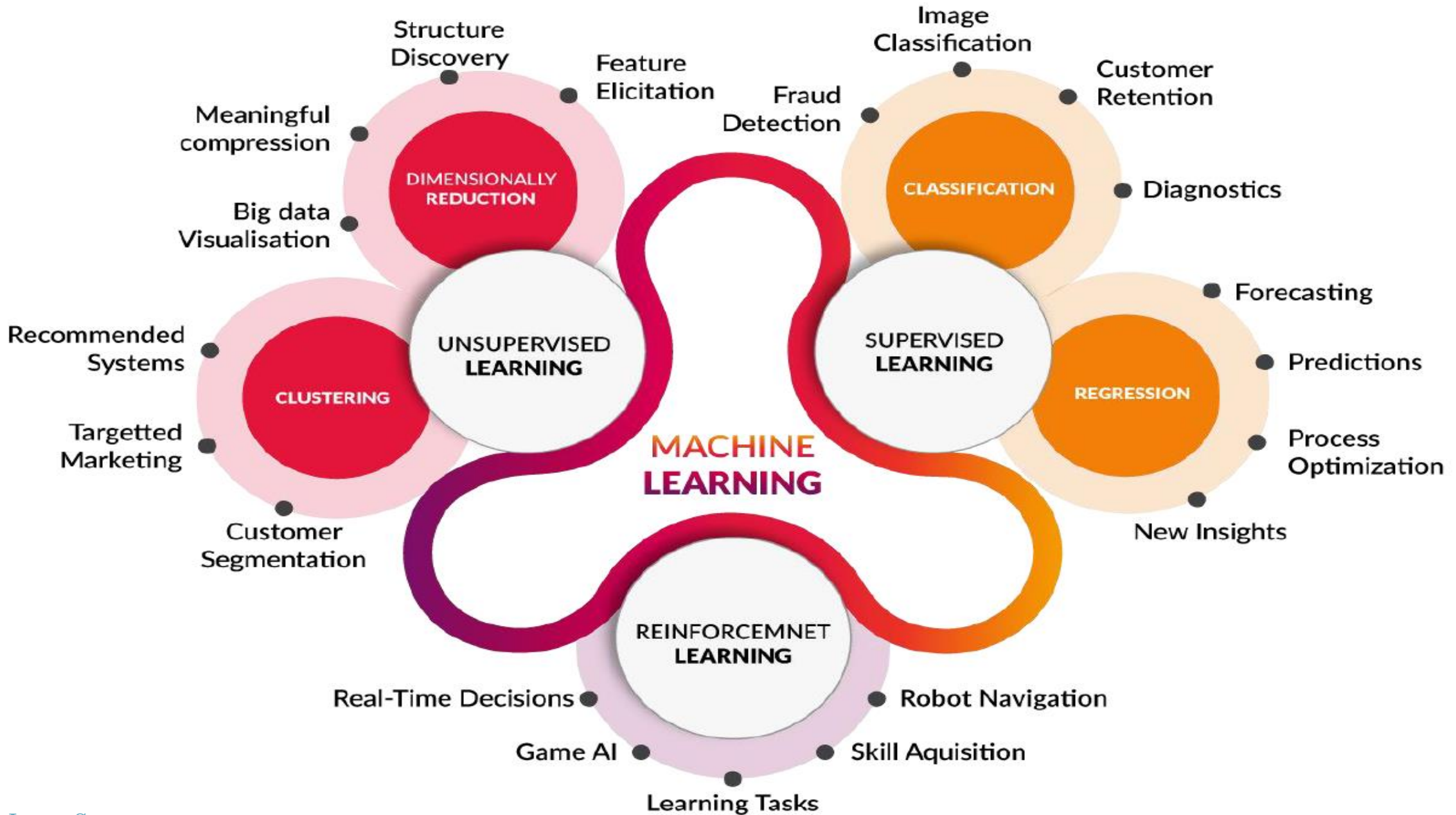
Big Data

Information Created Worldwide =
Expected to Continue Accelerating

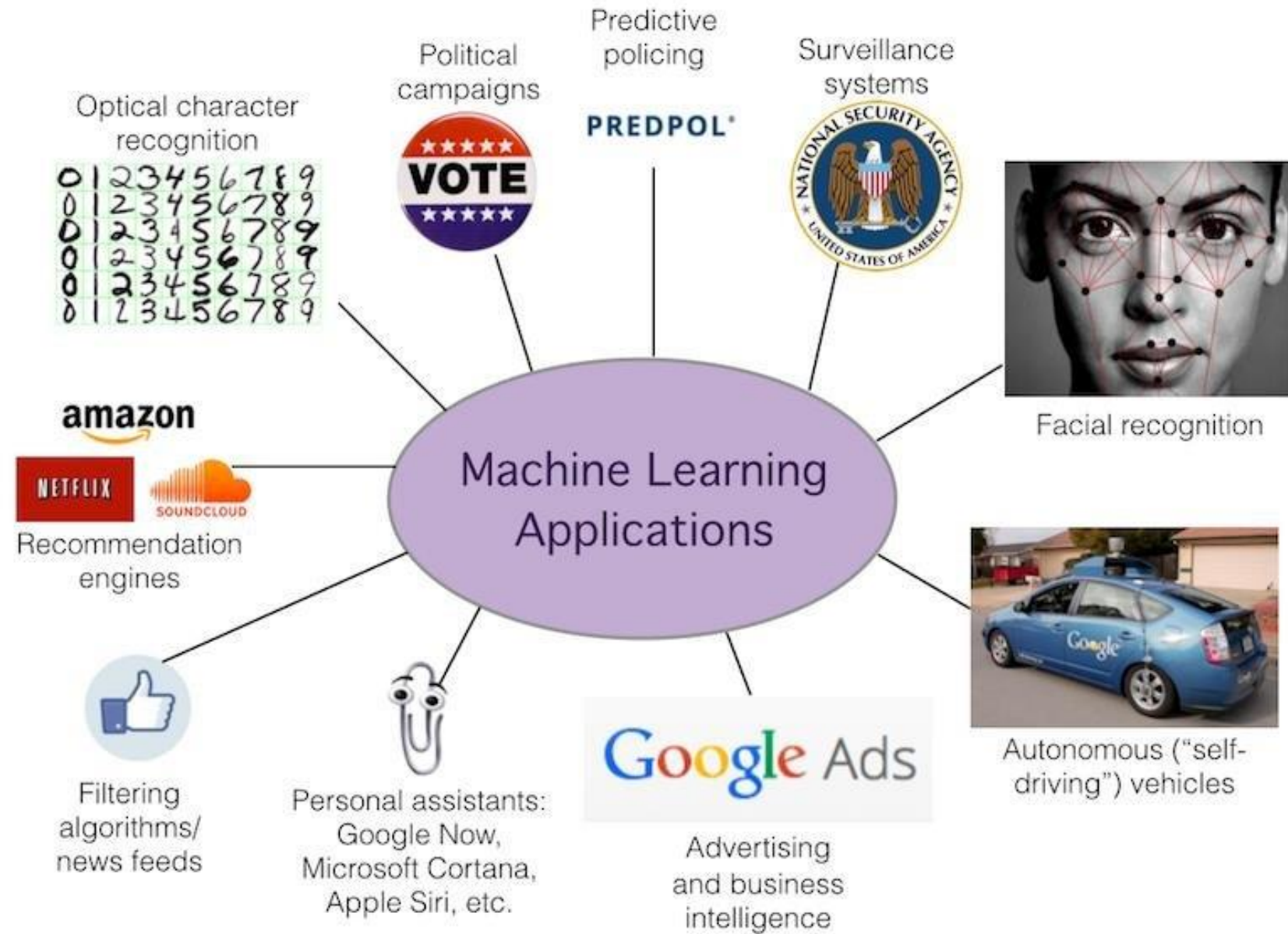


ML Applications

ML Applications



ML Applications



Recommender System

- Recommender Systems aim to help a user or a group of users to select items from a crowded item or information space.
- Types of Recommender Systems
 - Most Popular Items
 - Association and Market Basket Models
 - Content Filtering
 - Collaborative Filtering
 - Hybrid Models

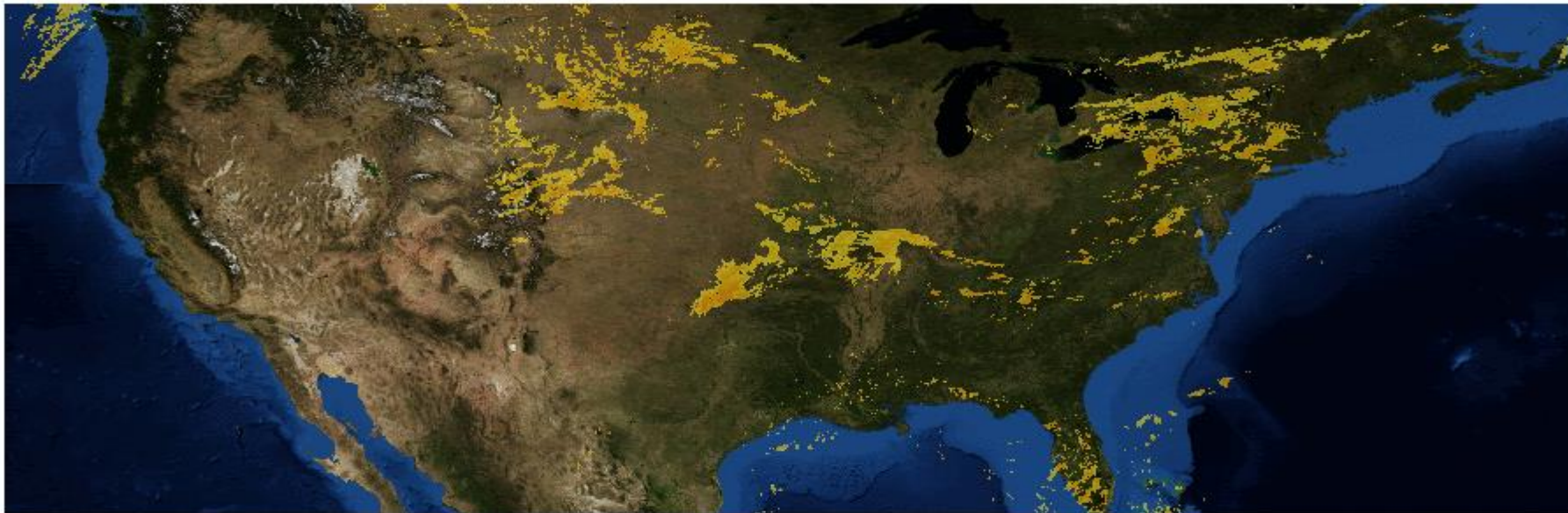
amazon.com

Recommended for You

Amazon.com has new recommendations for you based on [items](#) you purchased or told us you own.



Weather Forecasting



[Image Source](#)

Banking

- Several banks have replaced older statistical modeling approaches with machine learning techniques and in some cases, experienced:
 - 10 % increases in sales of new products
 - 20 % savings in capital expenditures
 - 20 % increases in cash collections
 - 20 % declines in churn.
- Devised new recommendation engines for clients in retailing and in small and medium-sized companies.
- Built microtargeted models that more accurately forecast who will cancel service or default on their loans, and how best to intervene.



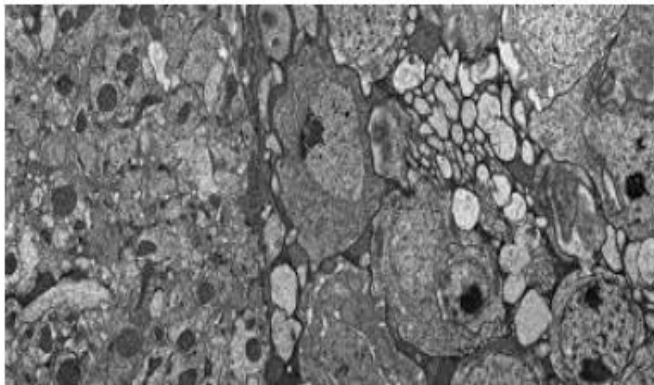
Computer Vision –Healthcare

Residual Deconvolutional Networks for Brain Electron Microscopy Image Segmentation

Ahmed Fakhry ; Tao Zeng ; Shuiwang Ji

Publication Year: 2017, Page(s):447 - 456

The automated 3D reconstruction of neurites in brain EM image stacks remains one of the most challenging problems in neuroscience [1]–[3]. In such problems, neurons spanning multiple adjacent image slices are expected to be consistently identified and reconstructed. Conventionally, this problem has been approached as a 2D prediction task, where each image slice is segmented individually. Then, a post-processing step was performed to generate 3D segmentation. The post-processing step usually involves heuristic off-the-shelf classifiers that were trained to link similar segments together across the entire image stack. These classifiers usually rely on hand-crafted features which incorporate prior knowledge and understanding of the data. Thus, classifiers that worked well on some problems/datasets are not guaranteed to perform similarly in different scenarios. It is thus desirable to design a fully trainable system with minimal post-processing to perform the 3D segmentation task in an end-to-end fashion.



<http://ieeexplore.ieee.org/document/7575638/>



Google Research Blog

The latest news from Research at Google

Deep Learning for Detection of Diabetic Eye Disease

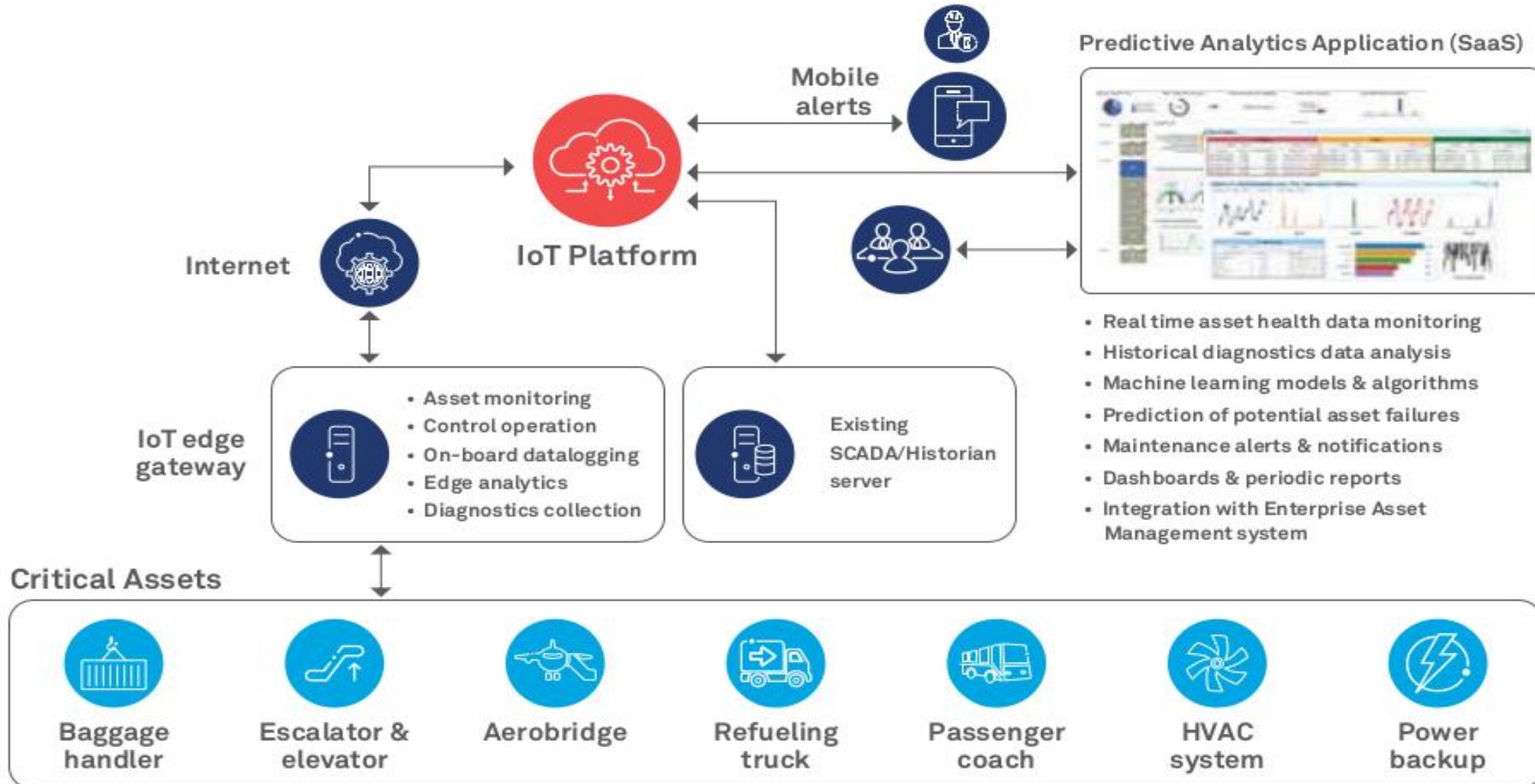
Tuesday, November 29, 2016

Posted by Lily Peng MD PhD, Product Manager and Varun Gulshan PhD, Research Engineer



<https://research.googleblog.com/2016/11/deep-learning-for-detection-of-diabetic.html>

Asset Management – Predictive Maintenance



Industrial AI

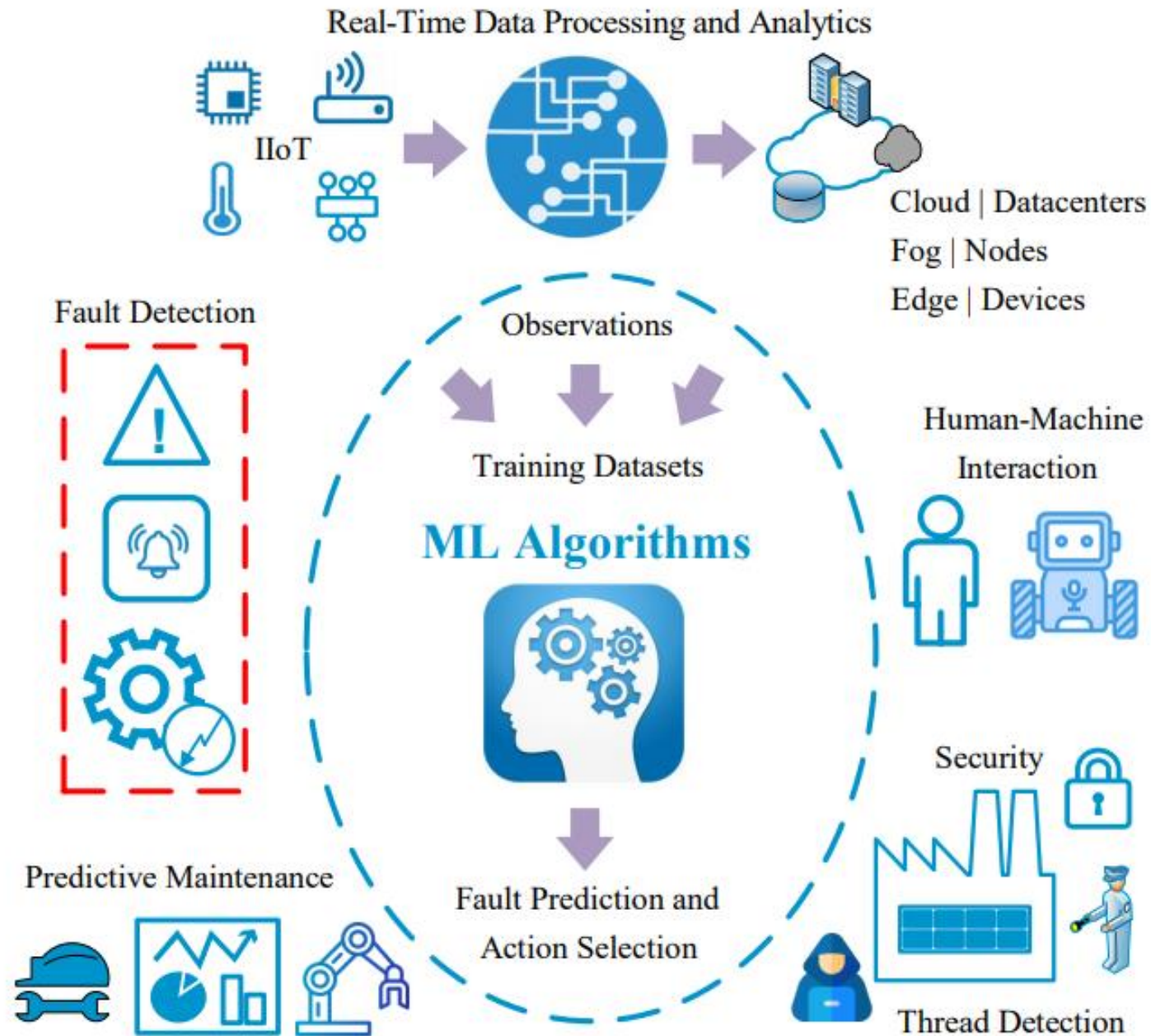


Image Source: Lee et al.

Few Quotes

- A breakthrough in machine learning would be worth ten Microsofts - **Bill Gates, Chairman, Microsoft**
- Machine learning is the next Internet - **Tony Tether, Director, DARPA**
- Machine learning is the hot new thing - **John Hennessy, President, Stanford**
- Web rankings today are mostly a matter of machine learning - **Prabhakar Raghavan, Dir. Research, Yahoo**
- Machine learning is going to result in a real revolution - **Greg Papadopoulos, CTO, Sun**
- Machine learning is today's discontinuity - **Jerry Yang, CEO, Yahoo**

Thank You
for
Attention

